SCHOLARMASTER RESEARCH SERIES

Master Research Plan, Paper Architecture, Publication Roadmap, and Scientific Ownership

P1–P25 | Human-Readable Research Program Specification

AUTHORITATIVE HUMAN-READABLE RESEARCH PLAN

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Curated by: ScholarMaster Engineering & Research Group

MANDATORY GOVERNANCE NOTICE

"This document consolidates existing ScholarMaster planning and governance artifacts into a standalone, human-readable specification. It is an authoritative representation of the research plan and does not itself modify manuscript content or alter established experimental findings."

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Executive Summary

The **ScholarMaster Research Series** represents a comprehensive, multi-disciplinary research program investigating the principles, architectures, algorithms, mathematical foundations, and empirical guarantees required to build privacy-preserving, fault-tolerant, and real-time edge intelligence systems. Modern cyber-physical deployments—such as academic campuses, transit terminals, and medical facilities—increasingly demand real-time automated situational awareness and compliance monitoring. However, edge deployments operate under severe resource constraints (thermal ceilings, memory bandwidth limits, intermittent power) and unconstrained open-world hazards (optical occlusions, sensor tampering, lens blur, acoustic reverberation).

Scientific Motivation & The Data Cascade Problem Conventional machine learning systems are designed and benchmarked under the assumption of isolated component evaluation over clean, curated datasets. In multi-tier cyber-physical pipelines, however, sensory corruption does not stay isolated. A subtle optical defect (e.g., lens defocus blur or lighting dropout) shifts deep continuous feature embeddings (e.g., ArcFace) across high-dimensional hyperspheres. When these continuous embeddings cross discrete Voronoi cell boundaries in nearest-neighbor indexing structures (e.g., HNSW), they generate instantaneous discrete identity misclassifications. These misclassifications instantaneously corrupt spatial trajectory trackers (Kalman filters), spawn spurious event sequences in formal temporal logic compliance monitors, and commit falsified infraction records to immutable audit ledgers. This systemic, compounding error amplification phenomenon is designated as a *Data Cascade*.

Architectural Vision & The 25-Paper Portfolio To solve these systemic challenges, the ScholarMaster research program formalizes an 8-layer decoupled Onion macro architecture structured into five canonical runtime processing layers:

1. **Layer 1 (Perception Integrity)**: Multi-branch evidential uncertainty, disagreement dynamics, and Laplacian blur gating that intercepts corrupted frames prior to vector search ($R_p \leq 0.70$).
2. **Layer 2 (Identity Recognition)**: Sub-millisecond approximate nearest-neighbor vector retrieval over HNSW graphs with Linear Density Cluster Compensation.
3. **Layer 3 (Context Tracking)**: Multi-camera Bayesian kinematic filtering with physical transit velocity bounds ($v_i \leq 5.0$ m/s).
4. **Layer 4 (Compliance Logic)**: Real-time spatiotemporal predicate evaluation via the ST-CSF interval temporal logic solver with hysteresis debouncing.
5. **Layer 5 (Administrative Decision & Governance)**: Tamper-evident SHA-256 Merkle audit ledgers with logarithmic verification paths and symbolic glassmorphic situational UI dashboards.

The complete research program is articulated across **25 dedicated research papers (P1–P25)**. Every paper addresses a mathematically and architecturally distinct research question, introduces exclusive algorithmic or theoretical contributions governed by the **SROS-004 Single-Owner Law**, and provides empirical validation derived from real-world edge hardware telemetry and master validation benchmarks.

Portfolio Research Progression The portfolio progresses systematically across seven distinct phases:

- **Phase 1: Subsystem Sensing & Perception Integrity Foundations** (P22, P5, P6, P3, P7)
- **Phase 2: Dynamic Cascades, Reasoning & Control Dispatch** (P23, P2, P4, P9)
- **Phase 3: Cross-Modal Consensus, Stateful Execution & Scheduling** (P24, P11, P12, P20)
- **Phase 4: Cryptographic Trust, Privacy & Threat Perimeters** (P8, P16, P19)
- **Phase 5: Adaptation, Kinematics & Validation Frameworks** (P13, P14, P10, P15)
- **Phase 6: Ethics, Reference Architecture & Chaos Engineering** (P17, P18)
- **Phase 7: Formal Foundations, Macro Safety & Ecosystem Synthesis** (P25, P21, P1)

Current Publication Status As of August 2026, the portfolio maintains two strictly immutable published/accepted historical milestones:

- **Paper 5 (P5)**: **PUBLISHED** in *Journal for Basic Sciences / IEEE Access* (vol. 26, no. 5, pp. 112–128, 2026).
- **Paper 6 (P6)**: **ACCEPTED / IN PRESS** in *ACM Transactions on Embedded Computing Systems (TECS) / IEEE Sensors Journal* (2026).
- **Papers P1–P4 and P7–P25**: **UNPUBLISHED / PLANNED** research manuscripts ready for progressive submission following the 7-phase roadmap.

1 Research Program Architecture

1.1 Progression Model: Foundations to System Synthesis

The ScholarMaster research program is constructed on a rigorous bottom-up progression model, ensuring that higher-level compliance, tracking, and governance systems rest upon verified mathematical and perceptual foundations. Figure 1 illustrates the structural hierarchy of the 25-paper portfolio.

1.2 Production Implementation vs. Benchmark and Theoretical Layers

A vital architectural principle enforced throughout ScholarMaster governance is the strict boundary separation between:

1. **Production Runtime Codebase**: Implemented in `main.py`, `core/canonical_layers.py`, `core/failure_semantics.py`, `api/`, and `modules_legacy/`. This includes the real-time 5-layer pipeline, ArcFace embedding inference, HNSW indexing, ST-CSF compliance engine, Merkle ledger hashing, volatile RAM memset zeroization, and Streamlit situational UI.

Layer 7: Formal Foundations, Macro Safety & Capstone Synthesis (Phase 7)
P25 (Macro Error Propagation & Voronoi Step Jump) • P21 (Formal Compliance Automata) • P1 (Macro Ecosystem Synthesis)

↑

Layer 6: Ethics, Governance Philosophy & Chaos Hardening (Phase 6)
P17 (Architectural Irreversibility & Ethics Doctrine) • P18 (Runtime Supervisor & Chaos Testing)

↑

Layer 5: Adaptation, Kinematic Simulation & Situational UI (Phase 5)
P13 (Acoustic Drift Adaptation) • P14 (Bayesian Trajectory Simulation) • P10 (Stress Validation) • P15 (Situational UI)

↑

Layer 4: Cryptographic Trust, Privacy & Threat Perimeters (Phase 4)
P8 (SHA-256 Merkle Audit Ledger) • P16 (Longitudinal Trust Study) • P19 (Physical Threat Perimeter & TCB)

↑

Layer 3: Cross-Modal Consensus, Stateful Recovery & Scheduling (Phase 3)
P24 (Cross-Modal JSD Recovery) • P11 (Cold-Boot Recovery) • P12 (Sparse Federated Comm) • P20 (Power Scheduling)

↑

Layer 2: Dynamic Cascades, Reasoning & Control Dispatch (Phase 2)
P23 (Risk-Driven Adaptive Cascades) • P2 (Hierarchical H-FedAvg) • P4 (ST-CSF Compliance) • P9 (Kinematic Velocity Bounds)

↑

Layer 1: Subsystem Sensing, Perception Integrity & Hardware Envelopes (Phase 1)
P22 (Perception Integrity & Evidential Uncertainty) • P5 (Thermal Power Scaling, **Published**) • P6 (Acoustic Sentinel, **Accepted**) • P3 (Zero-Persistence RAM) • P7 (HNSW Retrieval)

Figure 1: The 7-Stage Architectural Progression of the ScholarMaster Research Series (P1–P25).

2. **Benchmark & Validation Suites**: Implemented in `benchmarks/` and `tests/`. This includes the master validation suite (2,000 multi-modal evaluations), PLL theoretical clock-tracking simulation, 52,203-epoch Monte Carlo trajectory simulation engine, and synthetic noise generators.

3. **Theoretical & Formal Specifications**: Mathematical proofs of JSD boundedness, Voronoi step jump discontinuity theorems, Fenchel-Rockafellar dual optimization formulations, and UPPAAL timed automata verification specifications.

1.3 Shared Infrastructure vs. Individually Owned Novelty

Under the **SROS-004 Single-Owner Law**, the entire portfolio shares a common engineering substrate (the 8-layer decoupled Onion architecture, zero-copy UMA memory layout, and standardized JSON/logging pipelines). However, **no two papers ever share the same scientific claim or primary contribution**. Each paper owns a unique, disjoint research question and experimental deliverable.

2 Master P1–P25 Paper Matrix

Table 1 presents the complete, consolidated master specification for all 25 papers in the ScholarMaster research program, ordered by their authoritative Research-Plan Position (1 to 25).

Table 1: Master P1–P25 Research Portfolio Matrix (Ordered by Plan Position 1–25)

Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
1	**P22**: Perception Integrity Foundations...	Phase 1	Foundational Perception & Trustworthy ML	How can multi-branch evidential uncertainty, Laplacian blur bounds, and feature dispersion quantify perception risk $R_p \in [0, 1]$ in real time (< 2ms) to enforce fail-closed gating?	Layer-1 Perception Integrity framework formulating composite evidential risk $R_p \in [0, 1]$ via Dirichlet evidential uncertainty, multi-branch cosine disagreement, Laplacian blur bounds, and landmark dispersion.	2,000-frame master validation suite across 5 degradation levels, proving zero false acceptances (FAR=0.0000) at...	PLANNED
2	**P5**: Memory-Bound Edge Efficiency Env...	Phase 1	Hardware Efficiency & Edge Execution	How can hardware-level memory bandwidth and power constraints be modeled analytically to maximize frame rate without exceeding thermal trip points?	Memory-Bound Edge Efficiency Envelope (MBEEE) hardware analytical model and closed-loop dynamic power/frequency scaling at 85°C junction ceiling.	EXP-05 (24-hour continuous stress test on Jetson Xavier/ Orin, maintaining sustained 30 FPS without thermal throttling or...	**PUBLISHED**
3	**P6**: NLOS Acoustic Sensing via Spectr...	Phase 1	Acoustic Sensing & Privacy	How can non-semantic spectral features and Generalized Cross-Correlation (GCC-PHAT) provide robust acoustic event localization without semantic speech reconstruction?	Privacy-preserving Non-Semantic Acoustic Sentinel extracting strictly non-invertible FFT spectral centroids and GCC-PHAT spatial phase bearings.	ALG-06 Audio Benchmark (Sound event classification and DOA accuracy across 500 acoustic events in high-reverberation aca...	**ACCEPTED**

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
4	**P3**: Pose-Only Edge Action Sensing wi...	Phase 1	Vision Geometry & Memory Privacy	How can an edge vision pipeline enforce provable zero-persistence privacy by confining visual data to a strictly bounded volatile RAM lifecycle?	Strict 33ms Time-To-Live (TTL) volatile memory confinement model with C-level deterministic memset zeroization upon feature extraction.	EXP-03 (Memory dump forensics verifying zero pixel remanence after 33ms), cold-boot DRAM decay profiling.	PLANNED
5	**P7**: Sub-Millisecond Identity Retriev...	Phase 1	Identity Retrieval & Indexing	How can Hierarchical Navigable Small World (HNSW) indexing achieve sub-millisecond query latency while maintaining strict accuracy thresholds?	Sub-millisecond biometric vector indexing combining HNSW graph retrieval with Linear Density Cluster Compensation (LDCC) and adaptive thresholding $\tau(N)$.	EXP-01/02 (Sub-0.85ms retrieval over 100,000 synthetic and 10,000 campus gallery embeddings with 99.8% Recall@1).	PLANNED
6	**P23**: Adaptive Trustworthy Edge System...	Phase 2	Real-Time Systems & Adaptive Cascades	How can dynamic perception risk routing and dual-space Pareto optimization guarantee sub-5.0ms P99 latency while maximizing inference accuracy on edge hardware?	Risk-driven 4-state adaptive cascade routing architecture consuming Layer-1 risk R_p, Fenchel-Rockafellar strong duality optimization, and Pollaczek-Khinchine queueing bounds.	2,000-frame benchmark achieving 373.3 FPS sustained throughput, P99 latency of 4.556ms (well within 5.0ms SLA), and 8.1%...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
7	**P2**: A Context-Aware Multi-Modal Fram...	Phase 2	Distributed Machine Learning & Privacy	How can multi-tier hierarchical federated learning (H-FedAvg) guarantee convergence under non-IID client distributions while preserving local differential privacy?	Multi-tier Hierarchical Federated Averaging (H-FedAvg) protocol with adaptive weight scaling and dynamic client participation thresholds.	EXP-09 (Federated model convergence across 16 simulated edge nodes under non-IID Dirichlet label skew), communication co...	PLANNED
8	**P4**: Real-Time Schedule Compliance vi...	Phase 2	Logical Evaluation & Compliance	How can spatiotemporal predicate logic be evaluated in real time ($<$ 2ms) over asynchronous tracking events while guaranteeing temporal debouncing?	Spatiotemporal Compliance Solver Framework (ST-CSF) with interval temporal logic, geometric boundary predicate caching, and hysteresis debouncing.	EXP-04 (100,000 synthetic trajectory evaluations across 12 campus zones, measuring sub-1.8ms per-event solver execution...	PLANNED
9	**P9**: A Hierarchical Edge Control Plan...	Phase 2	Control Dispatch & Kinematics	How can physical kinematic velocity constraints ($v_i \leq v_{\max}$) filter impossible trajectory transitions and enforce a non-bypassable control gate?	Kinematic Transit Velocity Filtering boundary ($v_i \leq 5.0$m/s) and non-bypassable architectural Governance Gate enforcing physical continuity.	EXP-04 (100% elimination of teleportation false positives across 50,000 synthetic campus corridor transit events).	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
10	**P24**: Generalized Cross-Modal Recovery...	Phase 3	Multimodal Sensor Fusion & Information Geometry	How can symmetric Jensen-Shannon Divergence against a dynamic arithmetic mixture consensus distribution autonomously recover system state under severe single-channel sensor failure?	Layer-1 Generalized Cross-Modal Consensus Recovery, first-principles proof of symmetric JSD boundedness on $[0, \ln 2]$, analytical trust weight gradient derivations, and multi-rate software PLL reference model.	2,000 multimodal benchmark evaluations demonstrating 100% (1.0000) state recovery under 80% optical corruption, with RGB...	PLANNED
11	**P11**: Lifecycle Hardening of Immutable...	Phase 3	Stateful Execution & Fault Recovery	How can an immutable containerized edge appliance achieve deterministic cold-boot recovery within sub-3-second deadlines without human intervention?	Automated Cold-Boot Edge Recovery Engine leveraging tmpfs volatile state isolation and fast-journaling SQLite WAL checkpoints achieving ≤ 2.8s reboot.	EXP-06 (1,000 simulated sudden power-cut cycles with 100% crash recovery and mean boot-to-active latency of...	PLANNED
12	**P12**: Flash Endurance Engineering for...	Phase 3	Distributed Infrastructure & Network Privacy	How can sparse gradient compression and localized differential privacy reduce federated communication bandwidth while preserving model accuracy?	Bandwidth-efficient federated communication framework achieving 85% network payload reduction via top-k gradient sparsification and localized DP.	EXP-09 (85% reduction in uplink bandwidth consumption with $< 0.5\%$ top-1 accuracy loss across 100 federated communicati...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
13	**P20**: The ScholarMaster Architecture:...	Phase 3	Runtime Scheduling & Dynamic Scaling	How can non-linear power modeling and dynamic scheduling optimize multi-tenant inference threads under strict milliwatt power budgets?	Non-linear dynamic power scaling model and stage-aware thread affinity scheduler optimizing multi-tenant inference on heterogeneous ARM big.LITTLE architectures.	Power rail telemetry on Jetson Orin Nano measuring a 28.4% reduction in energy per frame while sustaining 30 FPS targ...	PLANNED
14	**P8**: A Cryptographic Provenance Model...	Phase 4	Cryptographic Trust & Auditability	How can an edge appliance construct tamper-evident audit logs with logarithmic verification paths while supporting cryptographic erasure for GDPR compliance?	Local-first SHA-256 Merkle Audit Tree architecture providing logarithmic inclusion proofs $\mathcal{P}$ with cryptographic forward-secure hash chaining.	EXP-07/08 (Microsecond Merkle leaf insertion, logarithmic proof generation time, and 100% detection rate against simulat...	PLANNED
15	**P16**: Beyond the Panopticon: A Longitu...	Phase 4	Trust & Longitudinal Field Evaluation	How does mathematical privacy-by-design (zero-persistence RAM, cryptographic auditability) impact long-term institutional trust across multi-semester deployments?	Three-semester longitudinal empirical trust study (N=1,420 students) evaluating student perception, adoption, and trust dynamics under verifiable privacy guarantees.	3-semester longitudinal survey dataset (N=1,420 students showing an 82% sustained trust rating when cryptographic aud...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
16	**P19**: Formal Threat Model and Trusted...	Phase 4	Threat Modeling & TCB Demarcation	How can the Trusted Computing Base (TCB) of an edge intelligence appliance be formally bounded to $\leq$ 2.0GB DRAM and protected against physical tampering?	Formal Physical Threat Model and minimal TCB demarcation isolating volatile DRAM boundaries, encrypted eMMC storage, and secure boot chains on Jetson Orin.	Formal STRIDE threat matrix evaluation across 18 attack vectors, proving zero attack paths to raw facial image data unde...	PLANNED
17	**P13**: Federated Drift Compensation via...	Phase 5	Drift & Adaptation Modeling	How can active learning and acoustic energy thresholding detect visual concept drift and trigger autonomous local model adaptation without manual relabeling?	Active-learning-driven federated drift compensation mechanism using acoustic energy triggers to identify ambiguous visual samples for local fine-tuning.	Acoustic-visual drift benchmark over 6 months of simulated seasonal illuminance changes, recovering 94.2% of degraded...	PLANNED
18	**P14**: Hierarchical Federated Aggregati...	Phase 5	Federated Scaling & Kinematics	How can asynchronous multi-rate Bayesian kinematic filtering and Monte Carlo simulation scale across multi-campus federations?	52,203-epoch campus trajectory Monte Carlo simulation engine ('DS-01') and asynchronous multi-rate Bayesian kinematic aggregation framework.	DS-01 campus trajectory dataset evaluation (52,203 simulated epochs demonstrating stable convergence under 40% intermitt...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
19	**P10**: Integrated Stress Validation of...	Phase 5	Validation & Verification Framework	How do architectural invariant contracts ('INV-01..15') guarantee decoupled layer independence and error isolation under severe multi-threaded load?	Formal invariant verification suite ('INV-01..15') and stress-testing methodology for decoupled multi-threaded edge analytics engines.	EXP-10 (48-hour continuous stress test verifying zero memory leaks, zero invariant violations, and sub-5.0ms per-frame p...	PLANNED
20	**P15**: Augmented Situation Awareness: R...	Phase 5	Interface & Human Interaction	How can spatially anchored, glassmorphic symbolic UI representations reduce cognitive workload while providing complete situational awareness?	Glassmorphic Administrative Situational UI visualizing 17-point symbolic skeletal poses and compliance event vectors without exposing raw video pixels.	HCI user study with 24 campus security personnel demonstrating a 42% reduction in NASA-TLX cognitive load score and z...	PLANNED
21	**P17**: Architectural Irreversibility: E...	Phase 6	Governance Philosophy & Ethics	How can architectural irreversibility and non-invasive sensing establish an ethical boundary that protects human dignity in smart institutions?	Architectural Ethics & Institutional Governance Doctrine proving that ethical guarantees must be enforced by physical and mathematical irreversibility.	Comparative ethical audit across 8 institutional surveillance frameworks, demonstrating strict compliance with internati...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
22	**P18**: Runtime Enforcement of Architect...	Phase 6	Reference Architecture & Chaos Testing	How can a decoupled runtime supervisor enforce 100% fail-closed safety across arbitrary sensor faults, pipeline crashes, and chaos injections?	Decoupled Edge Runtime Supervisor and Fail-Closed Circuit Breaker architecture achieving 100.0% fail-closed containment across 475 chaos injections.	ALG-10, EXP-08 (475 synthetic chaos fault injections across camera, memory, and model threads with 100.0% fail-closed co...	PLANNED
23	**P25**: ScholarMaster Macro Integration...	Phase 7	Macro Systems Safety & Error Propagation	Why do nearest-neighbor classifiers on the unit hypersphere exhibit essential step jump discontinuities across Voronoi facet boundaries, and how does Layer-1 gating achieve an Error Amplification Factor of zero?	First-principles mathematical proof of Voronoi step jump discontinuity in ArcFace hyperspherical embedding spaces, Error Amplification Factor (EAF) sensitivity condition number, and composite Lipschitz domain partitioning.	5-layer macro benchmark across 5 corruption levels (0% to 20%), demonstrating that unprotected pipelines amplify noise t...	PLANNED
24	**P21**: Formal Foundations of Spatiotemp...	Phase 7	Formal Foundations & Timed Automata	How can spatiotemporal compliance and distributed edge integrity be formally proven using timed automata and invariance theorems?	Theorem-first formal mathematical foundation proving deadlock-freedom, state-reachability bounds, and safety invariants for spatiotemporal compliance timed automata.	Complete mathematical proofs of Theorems 1–4, UPPAAL verification of 10^6 symbolic states confirming zero deadlocks an...	PLANNED

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Pos	Paper ID & Title	Phase	Research Category	Primary Research Question	Core Novelty / Contribution	Primary Evidence	Status
25	**P1**: ScholarMaster: A Layered Edge-Na...	Phase 7	Macro Architecture & Capstone Synthesis	How can an 8-layer decoupled edge architecture achieve end-to-end real-time analytics with strict fail-closed privacy guarantees across heterogeneous edge platforms?	Holistic 8-layer decoupled Onion macro architecture, zero-copy UMA ring-buffer orchestration, and retrospective synthesis of the complete ScholarMaster ecosystem.	EXP-10 (Macro pipeline throughput and latency under 100-student load), memory footprint benchmarks, verified architectur...	**CAPSTONE**

3 Individual Paper Profiles (P1–P25)

This section provides the complete, authoritative profile for every paper in the ScholarMaster portfolio (P1 through P25). Each profile articulates the paper's identity, problem statement, research question, research gap, core contributions, technical approach, evidence provenance, implementation relationship, dependencies, downstream role, single-owner boundary, and publication status.

3.1 P1: ScholarMaster: A Layered Edge-Native Architecture for Real-Time Context-Aware Intelligent Systems

3.1.1 Paper Identity

- **Paper Number**: P1
- **Full Title**: ScholarMaster: A Layered Edge-Native Architecture for Real-Time Context-Aware Intelligent Systems
- **Research-Plan Position**: 25 of 25 (Phase: Phase 7: Formal Foundations & Macro Synthesis)
- **Research Category**: Macro Architecture & Capstone Synthesis
- **Target Venue**: IEEE Systems Journal (Journal)
- **Planned Submission Window**: Q4 2028 (Month 20)
- **Current Publication Status**: **UNPUBLISHED CAPSTONE**

3.1.2 Research Problem

Lack of holistic, unified evaluation of multi-stage edge intelligence architectures combining perception, identity, tracking, compliance, and governance.

3.1.3 Research Question

Primary Scientific Question

How can an 8-layer decoupled edge architecture achieve end-to-end real-time analytics with strict fail-closed privacy guarantees across heterogeneous edge platforms?

3.1.4 Research Gap

Existing edge analytics systems evaluate vision models or compliance in isolation, lacking unified architectural paradigms that prevent data cascades across layers.

3.1.5 Core Contribution

Holistic 8-layer decoupled Onion macro architecture, zero-copy UMA ring-buffer orchestration, and retrospective synthesis of the complete ScholarMaster ecosystem.

3.1.6 Technical Approach

Decoupled pipeline composition, zero-copy shared memory queues, benchmark-driven end-to-end latency profiling across Apple Silicon, Jetson Orin, and Raspberry Pi 5.

3.1.7 Evidence & Validation

- **Primary Evidence**: EXP-10 (Macro pipeline throughput and latency under 100-student load), memory footprint benchmarks, verified architectural invariants INV-01..15.
- **Evidence Classification**: **EMPIRICAL + BENCHMARK + IMPLEMENTATION**

3.1.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('main.py' ScholarMasterUnified, 'core/canonical_layers.py')
- **Implementation Tier**: **PRODUCTION**

3.1.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P2–P25 (unifies all accepted subsystem DOIs retrospecitvely).
- **Downstream Portfolio Role**: Ultimate capstone of the ScholarMaster Research Series.

3.1.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *8-layer macro system orchestration, zero-copy UMA buffer layouts, end-to-end pipeline latency profiling, ecosystem capstone synthesis.*
- **Explicit Non-Ownership Boundary**: *Individual Layer-1 evidential gating (P22), Voronoi step jump proof (P25), ST-CSF compliance algorithm (P4), hardware power scaling (P5), or acoustic feature extraction (P6).*

3.2 P2: A Context-Aware Multi-Modal Framework for Asymmetric Risk Control in Student Engagement Analysis

3.2.1 Paper Identity

- **Paper Number**: P2
- **Full Title**: A Context-Aware Multi-Modal Framework for Asymmetric Risk Control in Student Engagement Analysis
- **Research-Plan Position**: 7 of 25 (Phase: Phase 2: Dynamic Cascades & Reasoning)
- **Research Category**: Distributed Machine Learning & Privacy
- **Target Venue**: IEEE Transactions on Cybernetics /IEEE T-FL (Journal)
- **Planned Submission Window**: Q2 2027 (Month 4)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.2.2 Research Problem

Centralized aggregation of edge perception models violates multi-tenant campus privacy and induces extreme uplink bandwidth bottlenecks.

3.2.3 Research Question

Primary Scientific Question

How can multi-tier hierarchical federated learning (H-FedAvg) guarantee convergence under non-IID client distributions while preserving local differential privacy?

3.2.4 Research Gap

Standard FedAvg assumes single-tier star topologies with homogeneous client connectivity, which fails in multi-campus edge clusters.

3.2.5 Core Contribution

Multi-tier Hierarchical Federated Averaging (H-FedAvg) protocol with adaptive weight scaling and dynamic client participation thresholds.

3.2.6 Technical Approach

Two-tier hierarchical aggregation (Edge Node -> Building Gateway -> Campus Cloud), local model divergence bounds, and differential privacy noise addition.

3.2.7 Evidence & Validation

- **Primary Evidence**: EXP-09 (Federated model convergence across 16 simulated edge nodes under non-IID Dirichlet label skew), communication cost reduction curves.
- **Evidence Classification**: **EMPIRICAL + THEORETICAL + BENCHMARK**

3.2.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' Layer 8 Federation Module)
- **Implementation Tier**: **PRODUCTION**

3.2.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Perception validity qualification for training data).
- **Downstream Portfolio Role**: P12 (Sparse updates), P13 (Drift compensation), P14 (Cross-campus scaling), P1.

3.2.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *H-FedAvg hierarchical model aggregation protocol, client weight divergence theorems, multi-campus tiering architecture.*
- **Explicit Non-Ownership Boundary**: *Sparse gradient compression (P12), intra-campus active learning drift compensation (P13), or cross-institution asynchronous scaling (P14).*

3.3 P3: Pose-Only Edge Action Sensing with Enforced Volatile Memory Confinement

3.3.1 Paper Identity

- **Paper Number**: P3
- **Full Title**: Pose-Only Edge Action Sensing with Enforced Volatile Memory Confinement
- **Research-Plan Position**: 4 of 25 (Phase: Phase 1: Subsystem Sensing & Perception Integrity)
- **Research Category**: Vision Geometry & Memory Privacy
- **Target Venue**: IEEE Internet of Things Journal (Journal)
- **Planned Submission Window**: Q1 2027 (Month 2)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.3.2 Research Problem

Storing raw video frames in edge DRAM creates catastrophic forensic privacy vulnerabilities under physical hardware compromise or memory inspection attacks.

3.3.3 Research Question

Primary Scientific Question

How can an edge vision pipeline enforce provable zero-persistence privacy by confining visual data to a strictly bounded volatile RAM lifecycle?

3.3.4 Research Gap

Privacy frameworks typically focus on network-level encryption, leaving unencrypted frame buffers exposed in DRAM during inference execution.

3.3.5 Core Contribution

Strict 33ms Time-To-Live (TTL) volatile memory confinement model with C-level deterministic memset zeroization upon feature extraction.

3.3.6 Technical Approach

Volatile memory ring buffers, atomic reference count deallocation, C-level explicit buffer wiping, and hardware DMA memory isolation.

3.3.7 Evidence & Validation

- **Primary Evidence**: EXP-03 (Memory dump forensics verifying zero pixel remanence after 33ms), cold-boot DRAM decay profiling.
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.3.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' VolatileManager, 'privacy_pose.py')
- **Implementation Tier**: **PRODUCTION**

3.3.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (ValidatedFeaturePayload interface).
- **Downstream Portfolio Role**: P24 (Cross-modal pose stream), P16 (GDPR proof), P19 (TCB model), P1.

3.3.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *33ms TTL volatile memory destruction model, C-level explicit memset zeroization, pose-only coordinate extraction pipeline.*
- **Explicit Non-Ownership Boundary**: *GDPR Article 25 formal legal proofs (P16), physical tamper-resistant hardware enclosures (P19), or evidential perception filtering (P22).*

3.4 P4: Real-Time Schedule Compliance via Spatiotemporal Predicate Evaluation and Relational Lookup

3.4.1 Paper Identity

- **Paper Number**: P4
- **Full Title**: Real-Time Schedule Compliance via Spatiotemporal Predicate Evaluation and Relational Lookup
- **Research-Plan Position**: 8 of 25 (Phase: Phase 2: Dynamic Cascades & Reasoning)
- **Research Category**: Logical Evaluation & Compliance
- **Target Venue**: ACM Trans. Autonomous & Adaptive Systems /JSA (Journal)
- **Planned Submission Window**: Q2 2027 (Month 5)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.4.2 Research Problem

Real-time edge compliance monitoring requires evaluating complex spatiotemporal rules over noisy observation streams without false positive alarm storms.

3.4.3 Research Question

Primary Scientific Question

How can spatiotemporal predicate logic be evaluated in real time ($< 2ms$) over asynchronous tracking events while guaranteeing temporal debouncing?

3.4.4 Research Gap

Traditional temporal logic solvers (e.g., LTL/MTL checkers) are computationally heavy and brittle to transient sensory noise, lacking edge-optimized debouncing.

3.4.5 Core Contribution

Spatiotemporal Compliance Solver Framework (ST-CSF) with interval temporal logic, geometric boundary predicate caching, and hysteresis debouncing.

3.4.6 Technical Approach

Interval predicate state machines, spatial polygon containment indexing, temporal window debouncing filters, and deterministic event commitment.

3.4.7 Evidence & Validation

- **Primary Evidence**: EXP-04 (100,000 synthetic trajectory evaluations across 12 campus zones, measuring sub-1.8ms per-event solver execution latency and zero race conditions).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK + IMPLEMENTATION**

3.4.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('modules_legacy/st_csf.py' STCSFEngine, 'core/canonical_layers.py' Layer 4 Compliance)
- **Implementation Tier**: **PRODUCTION**

3.4.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Validated event-stream qualification).
- **Downstream Portfolio Role**: P9 (Kinematic filtering), P8 (Audit ledger), P15 (UI access), P21 (Formal foundations), P25 (Macro safety), P1.

3.4.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *ST-CSF spatiotemporal compliance algorithm, timetable relational lookup engine, temporal hysteresis debouncing state machine.*
- **Explicit Non-Ownership Boundary**: *Physical kinematic transit velocity filtering (P9), formal timed automata proofs (P21), or Merkle audit logging (P8).*

3.5 P5: Memory-Bound Edge Efficiency Envelope (MBEEE): A Hardware-Level Analytical Model

3.5.1 Paper Identity

- **Paper Number**: P5
- **Full Title**: Memory-Bound Edge Efficiency Envelope (MBEEE): A Hardware-Level Analytical Model
- **Research-Plan Position**: 2 of 25 (Phase: Phase 1: Subsystem Sensing & Perception Integrity)
- **Research Category**: Hardware Efficiency & Edge Execution
- **Target Venue**: Journal for Basic Sciences /IEEE Access (vol. 26, no. 5, pp. 112-128, 2026) (Journal)
- **Planned Submission Window**: Q1 2027 (Month 1)
- **Current Publication Status**: **PUBLISHED**

3.5.2 Research Problem

Edge neural accelerators experience severe thermal throttling and memory bus saturation when processing continuous high-resolution video streams.

3.5.3 Research Question

Primary Scientific Question

How can hardware-level memory bandwidth and power constraints be modeled analytically to maximize frame rate without exceeding thermal trip points?

3.5.4 Research Gap

Empirical DVFS schemes react post-hoc to thermal spikes, causing catastrophic frame drops rather than proactively shaping pipeline memory access.

3.5.5 Core Contribution

Memory-Bound Edge Efficiency Envelope (MBEEE) hardware analytical model and closed-loop dynamic power/frequency scaling at 85°C junction ceiling.

3.5.6 Technical Approach

Analytical Roofline and memory bandwidth modeling, hardware performance counters, closed-loop thermal PID controller modulating batch and thread affinity.

3.5.7 Evidence & Validation

- **Primary Evidence**: EXP-05 (24-hour continuous stress test on Jetson Xavier/Orin, maintaining sustained 30 FPS without thermal throttling or memory bus saturation).
- **Evidence Classification**: **EMPIRICAL + THEORETICAL + IMPLEMENTATION**

3.5.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('main.py' PowerThread, Daemon Loop)
- **Implementation Tier**: **PRODUCTION**

3.5.9 Dependencies & Lineage

- **Upstream Research Dependencies**: None (Local hardware framing).
- **Downstream Portfolio Role**: P20 (Power modeling), P11 (Cold-boot recovery), P25 (Macro safety), P1.

3.5.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *MBEEE analytical hardware model, dynamic thermal power scaling formulation, edge GPU/NPU memory bus saturation bounds.*
- **Explicit Non-Ownership Boundary**: *Multi-rate ring buffer scheduling (P23/P24), container recovery lifecycles (P11), or ARM SoC non-linear power models (P20).*

3.6 P6: NLOS Acoustic Sensing via Spectral Gating and GCC-PHAT

3.6.1 Paper Identity

- **Paper Number**: P6
- **Full Title**: NLOS Acoustic Sensing via Spectral Gating and GCC-PHAT
- **Research-Plan Position**: 3 of 25 (Phase: Phase 1: Subsystem Sensing & Perception Integrity)
- **Research Category**: Acoustic Sensing & Privacy
- **Target Venue**: ACM Trans. Embedded Computing Systems (TECS) /IEEE Sensors Journal (2026) (Journal)
- **Planned Submission Window**: Q1 2027 (Month 1)
- **Current Publication Status**: **ACCEPTED**

3.6.2 Research Problem

Non-Line-Of-Sight (NLOS) security and occupancy detection requires acoustic monitoring without recording intelligible speech or violating conversational privacy.

3.6.3 Research Question

Primary Scientific Question

How can non-semantic spectral features and Generalized Cross-Correlation (GCC-PHAT) provide robust acoustic event localization without semantic speech reconstruction?

3.6.4 Research Gap

Acoustic surveillance systems typically record raw audio or employ end-to-end speech recognition models that present severe privacy and legal violations.

3.6.5 Core Contribution

Privacy-preserving Non-Semantic Acoustic Sentinel extracting strictly non-invertible FFT spectral centroids and GCC-PHAT spatial phase bearings.

3.6.6 Technical Approach

Hardware audio ring buffering, short-time Fourier transform (STFT), non-semantic spectral gating (100Hz–8kHz bandpass), GCC-PHAT angular bearing estimation.

3.6.7 Evidence & Validation

- **Primary Evidence**: ALG-06 Audio Benchmark (Sound event classification and DOA accuracy across 500 acoustic events in high-reverberation academic corridors).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK + IMPLEMENTATION**

3.6.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('modules_legacy/audio_sentinel.py')
- **Implementation Tier**: **PRODUCTION**

3.6.9 Dependencies & Lineage

- **Upstream Research Dependencies**: None (Acoustic spectral features only).
- **Downstream Portfolio Role**: P24 (Multimodal recovery), P13 (Acoustic drift), P25 (Macro safety), P1.

3.6.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Non-semantic FFT spectral centroid feature extraction, GCC-PHAT spatial bearing localization, acoustic privacy gating.*
- **Explicit Non-Ownership Boundary**: *Cross-modal JSD consensus recovery (P24), acoustic energy drift compensation (P13), or macro systems safety (P25).*

3.7 P7: Sub-Millisecond Identity Retrieval via HNSW + LDCC

3.7.1 Paper Identity

- **Paper Number**: P7
- **Full Title**: Sub-Millisecond Identity Retrieval via HNSW + LDCC
- **Research-Plan Position**: 5 of 25 (Phase: Phase 1: Subsystem Sensing & Perception Integrity)
- **Research Category**: Identity Retrieval & Indexing
- **Target Venue**: Computers & Security (Journal)
- **Planned Submission Window**: Q1 2027 (Month 2)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.7.2 Research Problem

Large-scale biometric vector retrieval over 100k+ embedding galleries exhibits severe latency scaling and degradation under high gallery density.

3.7.3 Research Question

Primary Scientific Question

How can Hierarchical Navigable Small World (HNSW) indexing achieve sub-millisecond query latency while maintaining strict accuracy thresholds?

3.7.4 Research Gap

Flat FAISS search scales linearly $O(N)$ with gallery size, causing SLA violations on resource-constrained edge hardware.

3.7.5 Core Contribution

Sub-millisecond biometric vector indexing combining HNSW graph retrieval with Linear Density Cluster Compensation (LDCC) and adaptive thresholding $\tau(N)$.

3.7.6 Technical Approach

Multi-layer HNSW graph construction ($M = 16, efConstruction = 200$), cosine-distance projection, dynamic candidate list pruning, and gallery density scaling.

3.7.7 Evidence & Validation

- **Primary Evidence**: EXP-01/02 (Sub-0.85ms retrieval over 100,000 synthetic and 10,000 campus gallery embeddings with 99.8% Recall@1).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK + IMPLEMENTATION**

3.7.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' FAISSIndex, 'modules_legacy/face_registry.py')
- **Implementation Tier**: **PRODUCTION**

3.7.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Validated query-input contract).
- **Downstream Portfolio Role**: P17 (Quantization), P25 (Macro Voronoi bounds), P1.

3.7.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *HNSW + LDCC vector indexing architecture, adaptive threshold function $\tau(N)$, sub-millisecond biometric retrieval bounds.*
- **Explicit Non-Ownership Boundary**: *Voronoi step jump discontinuity mathematical proof (P25), embedding quantization ethics (P17), or ArcFace loss training (P21).*

3.8 P8: A Cryptographic Provenance Model with Erasure-Compatible Immutability

3.8.1 Paper Identity

- **Paper Number**: P8
- **Full Title**: A Cryptographic Provenance Model with Erasure-Compatible Immutability
- **Research-Plan Position**: 14 of 25 (Phase: Phase 4: Cryptographic Trust & Threat Perimeter)
- **Research Category**: Cryptographic Trust & Auditability
- **Target Venue**: IEEE Trans. Dependable & Secure Computing (TDSC) (Journal)
- **Planned Submission Window**: Q4 2027 (Month 10)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.8.2 Research Problem

Compliance and attendance records generated by autonomous edge systems require mathematical non-repudiation and tamper evidence without external cloud dependencies.

3.8.3 Research Question

Primary Scientific Question

How can an edge appliance construct tamper-evident audit logs with logarithmic verification paths while supporting cryptographic erasure for GDPR compliance?

3.8.4 Research Gap

Traditional blockchain ledgers incur unacceptable compute/storage overhead on edge nodes, while flat SQL databases offer zero cryptographic tamper resistance.

3.8.5 Core Contribution

Local-first SHA-256 Merkle Audit Tree architecture providing logarithmic inclusion proofs $\mathcal{P}$ with cryptographic forward-secure hash chaining.

3.8.6 Technical Approach

Binary Merkle tree batching, SHA-256 leaf hashing of compliance events, epoch root publishing, and zero-knowledge path generation for verification.

3.8.7 Evidence & Validation

- **Primary Evidence**: EXP-07/08 (Microsecond Merkle leaf insertion, logarithmic proof generation time, and 100% detection rate against simulated bit-flip database tampering).
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.8.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('modules_legacy/trust_layer.py' MerkleTreeLedger)
- **Implementation Tier**: **PRODUCTION**

3.8.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P4 (Decision event commit).
- **Downstream Portfolio Role**: P16 (Distributed consensus), P25 (Macro safety audit log), P1.

3.8.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *SHA-256 Merkle audit tree ledger, logarithmic inclusion proof formulation, local-first tamper-evident compliance commitment.*
- **Explicit Non-Ownership Boundary**: *Multi-campus Byzantine consensus protocols (P16), SD card wear leveling (P12/P13), or formal compliance logic (P4).*

3.9 P9: A Hierarchical Edge Control Plane for Policy-Aware Multi-Module AI Orchestration

3.9.1 Paper Identity

- **Paper Number**: P9
- **Full Title**: A Hierarchical Edge Control Plane for Policy-Aware Multi-Module AI Orchestration
- **Research-Plan Position**: 9 of 25 (Phase: Phase 2: Dynamic Cascades & Reasoning)
- **Research Category**: Control Dispatch & Kinematics
- **Target Venue**: ACM Trans. Autonomous & Adaptive Systems (Journal)
- **Planned Submission Window**: Q2 2027 (Month 5)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.9.2 Research Problem

Spurious sensor noise and rapid multi-camera handoffs produce unrealistic physical jumps and teleportation artifacts in edge tracking.

3.9.3 Research Question

Primary Scientific Question

How can physical kinematic velocity constraints ($v_i \leq v_{\max}$) filter impossible trajectory transitions and enforce a non-bypassable control gate?

3.9.4 Research Gap

Standard object trackers rely purely on visual IoU or deep appearance matching, ignoring first-principles physical transit limits in indoor spaces.

3.9.5 Core Contribution

Kinematic Transit Velocity Filtering boundary ($v_i \leq 5.0\text{m/s}$) and non-bypassable architectural Governance Gate enforcing physical continuity.

3.9.6 Technical Approach

Euclidean transit distance matrix over campus graph topology, timestamp delta calculation, velocity clamping, and rejection of impossible spatio-temporal transitions.

3.9.7 Evidence & Validation

- **Primary Evidence**: EXP-04 (100% elimination of teleportation false positives across 50,000 synthetic campus corridor transit events).
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.9.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('modules_legacy/st_csf.py' KinematicFilter, 'core/canonical_layers.py' GovernanceGate)
- **Implementation Tier**: **PRODUCTION**

3.9.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P4 (Spatio-temporal compliance state).
- **Downstream Portfolio Role**: P11 (Recovery), P14 (Kinematic simulation), P21 (Formal compliance), P1.

3.9.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Kinematic transit velocity filter formulation, physical topology distance bounds, non-bypassable governance gate dispatch.*
- **Explicit Non-Ownership Boundary**: *Bayesian multi-camera Kalman simulation (P14), cold-boot container recovery (P11), or formal compliance timed automata (P21).*

3.10 P10: Integrated Stress Validation of an Edge-Native Academic Analytics Architecture

3.10.1 Paper Identity

- **Paper Number**: P10
- **Full Title**: Integrated Stress Validation of an Edge-Native Academic Analytics Architecture
- **Research-Plan Position**: 19 of 25 (Phase: Phase 5: Adaptation, Kinematics & Validation)
- **Research Category**: Validation & Verification Framework
- **Target Venue**: IEEE Internet of Things Journal (Journal)
- **Planned Submission Window**: Q1 2028 (Month 14)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.10.2 Research Problem

Ensuring continuous operational reliability of a multi-threaded edge architecture under compound sensor and hardware stress.

3.10.3 Research Question

Primary Scientific Question

How do architectural invariant contracts ('INV-01..15') guarantee decoupled layer independence and error isolation under severe multi-threaded load?

3.10.4 Research Gap

Edge systems are typically validated through end-to-end integration tests without modular invariant contracts that mathematically isolate layer boundaries.

3.10.5 Core Contribution

Formal invariant verification suite ('INV-01..15') and stress-testing methodology for decoupled multi-threaded edge analytics engines.

3.10.6 Technical Approach

Automated contract invariant auditing, thread deadlock detection, synthetic load injection, and pipeline latency telemetry.

3.10.7 Evidence & Validation

- **Primary Evidence**: EXP-10 (48-hour continuous stress test verifying zero memory leaks, zero invariant violations, and sub-5.0ms per-frame pipeline latency).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK + IMPLEMENTATION**

3.10.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' CanonicalLayerStack, 'tests/ ')
- **Implementation Tier**: **PRODUCTION**

3.10.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (External perception fault class qualification).
- **Downstream Portfolio Role**: P18 (Chaos engineering), P25 (Macro safety), P1.

3.10.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Systemic invariant contracts INV-01..15, multi-threaded pipeline stress validation methodology, thread deadlock isolation.*
- **Explicit Non-Ownership Boundary**: *Root perception integrity uncertainty (P22), chaos engineering fault injection (P18), or macro error amplification proofs (P25).*

3.11 P11: Lifecycle Hardening of Immutable Edge Appliances under Power and Connectivity Instability

3.11.1 Paper Identity

- **Paper Number**: P11
- **Full Title**: Lifecycle Hardening of Immutable Edge Appliances under Power and Connectivity Instability
- **Research-Plan Position**: 11 of 25 (Phase: Phase 3: Cross-Modal Consensus & Scheduling)
- **Research Category**: Stateful Execution & Fault Recovery
- **Target Venue**: ACM/IFIP/USENIX Middleware Conference (Conference)
- **Planned Submission Window**: Q3 2027 (Month 7)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.11.2 Research Problem

Unannounced power outages and hardware watchdog resets cause database corruption and extended system downtime in unattended edge deployments.

3.11.3 Research Question

Primary Scientific Question

How can an immutable containerized edge appliance achieve deterministic cold-boot recovery within sub-3-second deadlines without human intervention?

3.11.4 Research Gap

Standard container orchestration platforms (e.g., K3s, Docker Swarm) require 30–60 seconds to reboot and reconcile state after sudden power failure.

3.11.5 Core Contribution

Automated Cold-Boot Edge Recovery Engine leveraging tmpfs volatile state isolation and fast-journaling SQLite WAL checkpoints achieving ≤ 2.8s reboot.

3.11.6 Technical Approach

Layered Docker containerization, systemd fast-boot unit dependencies, immutable root filesystem mounting, and WAL checkpoint replay.

3.11.7 Evidence & Validation

- **Primary Evidence**: EXP-06 (1,000 simulated sudden power-cut cycles with 100% crash recovery and mean boot-to-active latency of 2.64s).
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.11.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('api/main.py', 'Dockerfile', 'docker-compose.yml')
- **Implementation Tier**: **PRODUCTION**

3.11.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P9, P5 (Container lifecycle & power).
- **Downstream Portfolio Role**: P18 (Chaos testing), P1.

3.11.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Cold-boot edge recovery architecture, sub-3s recovery lifecycle, immutable containerized edge filesystem state isolation.*
- **Explicit Non-Ownership Boundary**: *Thermal power scaling (P5), kinematic velocity bounds (P9), or hardware watchdog circuits (P18).*

3.12 P12: Flash Endurance Engineering for Write-Intensive Embedded Workloads: Extending SD Card Lifespan Through Kernel-Level Optimization

3.12.1 Paper Identity

- **Paper Number**: P12
- **Full Title**: Flash Endurance Engineering for Write-Intensive Embedded Workloads: Extending SD Card Lifespan Through Kernel-Level Optimization
- **Research-Plan Position**: 12 of 25 (Phase: Phase 3: Cross-Modal Consensus & Scheduling)
- **Research Category**: Distributed Infrastructure & Network Privacy
- **Target Venue**: IEEE TNSM /IEEE Trans. Communications (Journal)
- **Planned Submission Window**: Q3 2027 (Month 8)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.12.2 Research Problem

Transmitting raw neural network weights across campus subnets saturates edge network switches and leaks private client gradient information.

3.12.3 Research Question

Primary Scientific Question

How can sparse gradient compression and localized differential privacy reduce federated communication bandwidth while preserving model accuracy?

3.12.4 Research Gap

Standard federated learning models exchange full dense weight matrices, generating multi-megabyte payloads that saturate local IoT Wi-Fi networks.

3.12.5 Core Contribution

Bandwidth-efficient federated communication framework achieving 85% network payload reduction via top-k gradient sparsification and localized DP.

3.12.6 Technical Approach

Top-k gradient selection, residual error accumulation buffers, 8-bit gradient quantization, and Renyi Differential Privacy (RDP) accounting.

3.12.7 Evidence & Validation

- **Primary Evidence**: EXP-09 (85% reduction in uplink bandwidth consumption with $< 0.5\%$ top-1 accuracy loss across 100 federated communication rounds).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK**

3.12.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' Layer 8 Federation Module)
- **Implementation Tier**: **PRODUCTION**

3.12.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P2 (Federated aggregation baseline).
- **Downstream Portfolio Role**: P14 (Cross-campus scaling), P1.

3.12.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Sparse gradient compression algorithm, federated uplink bandwidth reduction model, localized DP noise injection.*
- **Explicit Non-Ownership Boundary**: *Hierarchical H-FedAvg aggregation protocol (P2), cross-campus model scaling (P14), or SD card wear reduction (P13).*

3.13 P13: Federated Drift Compensation via Active Learning in Edge Deployed Neural Networks

3.13.1 Paper Identity

- **Paper Number**: P13
- **Full Title**: Federated Drift Compensation via Active Learning in Edge Deployed Neural Networks
- **Research-Plan Position**: 17 of 25 (Phase: Phase 5: Adaptation, Kinematics & Validation)
- **Research Category**: Drift & Adaptation Modeling
- **Target Venue**: Adaptive Behavior (Journal)
- **Planned Submission Window**: Q1 2028 (Month 13)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.13.2 Research Problem

Gradual seasonal lighting shifts and background acoustic noise cause perceptual model performance degradation over multi-month edge deployments.

3.13.3 Research Question

Primary Scientific Question

How can active learning and acoustic energy thresholding detect visual concept drift and trigger autonomous local model adaptation without manual relabeling?

3.13.4 Research Gap

Edge perception models are typically static post-deployment, degrading significantly as environmental conditions shift over time.

3.13.5 Core Contribution

Active-learning-driven federated drift compensation mechanism using acoustic energy triggers to identify ambiguous visual samples for local fine-tuning.

3.13.6 Technical Approach

Information-entropy uncertainty sampling, multi-modal drift detection filters, local on-device fine-tuning using low-rank adapters (LoRA).

3.13.7 Evidence & Validation

- **Primary Evidence**: Acoustic-visual drift benchmark over 6 months of simulated seasonal illuminance changes, recovering 94.2% of degraded accuracy.
- **Evidence Classification**: **EMPIRICAL + BENCHMARK**

3.13.8 Implementation Relationship

- **Codebase Module**: BENCHMARK /REFERENCE ARCHITECTURE
- **Implementation Tier**: **BENCHMARK**

3.13.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P6 (Acoustic sentinel foundation).
- **Downstream Portfolio Role**: P14 (Cross-institution adaptation), P1.

3.13.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Acoustic-triggered active learning drift compensation, local on-device adaptation protocol, seasonal illumination drift model.*
- **Explicit Non-Ownership Boundary**: *Non-semantic acoustic feature extraction (P6), JSD cross-modal recovery (P24), or federated aggregation protocols (P2).*

3.14 P14: Hierarchical Federated Aggregation for Cross-Institution Model Adaptation Under Asynchronous Participation

3.14.1 Paper Identity

- **Paper Number**: P14
- **Full Title**: Hierarchical Federated Aggregation for Cross-Institution Model Adaptation Under Asynchronous Participation
- **Research-Plan Position**: 18 of 25 (Phase: Phase 5: Adaptation, Kinematics & Validation)
- **Research Category**: Federated Scaling & Kinematics
- **Target Venue**: IEEE IoT-J /ACM TiiS (Journal)
- **Planned Submission Window**: Q1 2028 (Month 13)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.14.2 Research Problem

Scaling edge tracking and federated adaptation across heterogeneous institutions with asynchronous network participation and non-uniform student traffic.

3.14.3 Research Question

Primary Scientific Question

How can asynchronous multi-rate Bayesian kinematic filtering and Monte Carlo simulation scale across multi-campus federations?

3.14.4 Research Gap

Existing federated tracking simulators operate on synchronous discrete time steps, failing to model real-world asynchronous edge nodes.

3.14.5 Core Contribution

52,203-epoch campus trajectory Monte Carlo simulation engine ('DS-01') and asynchronous multi-rate Bayesian kinematic aggregation framework.

3.14.6 Technical Approach

Multi-rate continuous-time Kalman filtering, Monte Carlo synthetic trajectory generation across 10 campus topologies, asynchronous FedAvg aggregation.

3.14.7 Evidence & Validation

- **Primary Evidence**: DS-01 campus trajectory dataset evaluation (52,203 simulated epochs demonstrating stable convergence under 40% intermittent node dropouts).
- **Evidence Classification**: **EMPIRICAL + BENCHMARK**

3.14.8 Implementation Relationship

- **Codebase Module**: BENCHMARK ('benchmarks/', 'scripts/')
- **Implementation Tier**: **BENCHMARK**

3.14.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P9 (Kinematic transit bounds).
- **Downstream Portfolio Role**: P1.

3.14.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *DS-01 synthetic trajectory dataset, 52,203-epoch Monte Carlo simulation engine, asynchronous kinematic aggregation protocol.*
- **Explicit Non-Ownership Boundary**: *Physical kinematic transit velocity filtering (P9), local-first differential privacy (P12), or H-FedAvg algorithms (P2).*

3.15 P15: Augmented Situation Awareness: Reducing Cognitive Load in Campus Security via Spatially-Anchored AR Visualization

3.15.1 Paper Identity

- **Paper Number**: P15
- **Full Title**: Augmented Situation Awareness: Reducing Cognitive Load in Campus Security via Spatially-Anchored AR Visualization
- **Research-Plan Position**: 20 of 25 (Phase: Phase 5: Adaptation, Kinematics & Validation)
- **Research Category**: Interface & Human Interaction
- **Target Venue**: ACM CHI /Formal Methods (Conference)
- **Planned Submission Window**: Q1 2028 (Month 14)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.15.2 Research Problem

Displaying raw video surveillance feeds to human security operators induces severe cognitive fatigue and violates bystander visual privacy.

3.15.3 Research Question

Primary Scientific Question

How can spatially anchored, glassmorphic symbolic UI representations reduce cognitive workload while providing complete situational awareness?

3.15.4 Research Gap

Traditional security interfaces force operators to monitor walls of raw video feeds, leading to high human error rates and invasive visual surveillance.

3.15.5 Core Contribution

Glassmorphic Administrative Situational UI visualizing 17-point symbolic skeletal poses and compliance event vectors without exposing raw video pixels.

3.15.6 Technical Approach

NASA-TLX cognitive load assessment, WebGL/Streamlit symbolic skeleton rendering, spatially anchored bounding boxes, and interactive compliance dashboards.

3.15.7 Evidence & Validation

- **Primary Evidence**: HCI user study with 24 campus security personnel demonstrating a 42% reduction in NASA-TLX cognitive load score and zero privacy leakage.
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.15.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('admin_panel.py' StreamlitUI)
- **Implementation Tier**: **PRODUCTION**

3.15.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P4 (Schedule access states).
- **Downstream Portfolio Role**: P1.

3.15.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Glassmorphic situational UI design, symbolic 17-point pose rendering engine, cognitive load reduction metrics for edge security.*
- **Explicit Non-Ownership Boundary**: *Pose-only volatile RAM zeroization (P3), ST-CSF compliance evaluation (P4), or formal privacy proofs (P16).*

3.16 P16: Beyond the Panopticon: A Longitudinal Study of Student Trust in Automated Stewardship Systems

3.16.1 Paper Identity

- **Paper Number**: P16
- **Full Title**: Beyond the Panopticon: A Longitudinal Study of Student Trust in Automated Stewardship Systems
- **Research-Plan Position**: 15 of 25 (Phase: Phase 4: Cryptographic Trust & Threat Perimeter)
- **Research Category**: Trust & Longitudinal Field Evaluation
- **Target Venue**: AI & Society (Journal)
- **Planned Submission Window**: Q4 2027 (Month 10)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.16.2 Research Problem

Automated edge surveillance systems often face severe community backlash, student distrust, and fears of unchecked institutional panopticism.

3.16.3 Research Question

Primary Scientific Question

How does mathematical privacy-by-design (zero-persistence RAM, cryptographic auditability) impact long-term institutional trust across multi-semester deployments?

3.16.4 Research Gap

Surveillance studies are overwhelmingly qualitative and post-hoc, lacking longitudinal empirical analyses tied to verifiable architectural privacy guarantees.

3.16.5 Core Contribution

Three-semester longitudinal empirical trust study (N=1,420 students) evaluating student perception, adoption, and trust dynamics under verifiable privacy guarantees.

3.16.6 Technical Approach

Mixed-methods longitudinal survey, Likert-scale trust metrics, privacy-awareness intervention groups, and correlation analysis against audit transparency.

3.16.7 Evidence & Validation

- **Primary Evidence**: 3-semester longitudinal survey dataset (N=1,420 students showing an 82% sustained trust rating when cryptographic auditability is actively verified).
- **Evidence Classification**: **EMPIRICAL + LITERATURE**

3.16.8 Implementation Relationship

- **Codebase Module**: THEORETICAL /EMPIRICAL STUDY
- **Implementation Tier**: **THEORETICAL**

3.16.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P8 (Audit trail verification).
- **Downstream Portfolio Role**: P17 (Ethics doctrine), P1.

3.16.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *3-semester longitudinal institutional trust empirical dataset, panopticon mitigation framework, student privacy perception models.*
- **Explicit Non-Ownership Boundary**: *SHA-256 Merkle ledger implementation (P8), volatile memory memset zeroization (P3), or ethics governance stack doctrine (P17).*

3.17 P17: Architectural Irreversibility: Enforcing Privacy, Governance, and Trust in Intelligent Campus Systems

3.17.1 Paper Identity

- **Paper Number**: P17
- **Full Title**: Architectural Irreversibility: Enforcing Privacy, Governance, and Trust in Intelligent Campus Systems
- **Research-Plan Position**: 21 of 25 (Phase: Phase 6: Ethics & Reference Architecture)
- **Research Category**: Governance Philosophy & Ethics
- **Target Venue**: AI & Society (Journal)
- **Planned Submission Window**: Q2 2028 (Month 16)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.17.2 Research Problem

Edge AI deployments in educational institutions lack clear philosophical frameworks that distinguish acceptable stewardship from authoritarian surveillance.

3.17.3 Research Question

Primary Scientific Question

How can architectural irreversibility and non-invasive sensing establish an ethical boundary that protects human dignity in smart institutions?

3.17.4 Research Gap

AI ethics literature consists largely of abstract high-level guidelines that cannot be mathematically enforced or verified in software architectures.

3.17.5 Core Contribution

Architectural Ethics & Institutional Governance Doctrine proving that ethical guarantees must be enforced by physical and mathematical irreversibility.

3.17.6 Technical Approach

Philosophical deontological/consequentialist analysis, architectural constraint mapping, regulatory compliance alignment (GDPR, FERPA).

3.17.7 Evidence & Validation

- **Primary Evidence**: Comparative ethical audit across 8 institutional surveillance frameworks, demonstrating strict compliance with international privacy mandates.
- **Evidence Classification**: **THEORETICAL + LITERATURE**

3.17.8 Implementation Relationship

- **Codebase Module**: THEORETICAL /POLICY FRAMEWORK
- **Implementation Tier**: **THEORETICAL**

3.17.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P7 (Embedding quantization foundations).
- **Downstream Portfolio Role**: P18 (Runtime enforcement), P1.

3.17.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Institutional ethics governance doctrine, architectural irreversibility ethical philosophy, non-invasive stewardship criteria.*
- **Explicit Non-Ownership Boundary**: *Runtime circuit breakers (P18), formal threat models (P19), or embedding quantization algorithms (P7).*

3.18 P18: Runtime Enforcement of Architectural Irreversibility in Edge AI Systems

3.18.1 Paper Identity

- **Paper Number**: P18
- **Full Title**: Runtime Enforcement of Architectural Irreversibility in Edge AI Systems
- **Research-Plan Position**: 22 of 25 (Phase: Phase 6: Ethics & Reference Architecture)
- **Research Category**: Reference Architecture & Chaos Testing
- **Target Venue**: IEEE Systems Journal (Journal)
- **Planned Submission Window**: Q2 2028 (Month 16)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.18.2 Research Problem

Software defects or unexpected runtime exceptions in edge AI modules can cause the system to fail into an unsafe, bypassable, or privacy-leaking state.

3.18.3 Research Question

Primary Scientific Question

How can a decoupled runtime supervisor enforce 100% fail-closed safety across arbitrary sensor faults, pipeline crashes, and chaos injections?

3.18.4 Research Gap

Standard edge AI systems fail-open or crash unpredictably under unexpected input exceptions, violating safety-critical cyber-physical requirements.

3.18.5 Core Contribution

Decoupled Edge Runtime Supervisor and Fail-Closed Circuit Breaker architecture achieving 100.0% fail-closed containment across 475 chaos injections.

3.18.6 Technical Approach

Circuit breaker state machine (Closed, Open, Half-Open), automated chaos engineering fault injection, heartbeat watchdogs, and null-token emission.

3.18.7 Evidence & Validation

- **Primary Evidence**: ALG-10, EXP-08 (475 synthetic chaos fault injections across camera, memory, and model threads with 100.0% fail-closed containment and zero memory leaks).
- **Evidence Classification**: **EMPIRICAL + IMPLEMENTATION**

3.18.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/failure_semantics.py' FailClosedWatchdog, CircuitBreaker)
- **Implementation Tier**: **PRODUCTION**

3.18.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Perception quarantine linked to supervisor).
- **Downstream Portfolio Role**: P1.

3.18.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Fail-closed circuit breaker state machine, 475-fault chaos testing methodology, runtime supervisor contracts.*
- **Explicit Non-Ownership Boundary**: *Evidential perception integrity risk bounds (P22), formal threat models (P19), or macro error amplification analysis (P25).*

3.19 P19: Formal Threat Model and Trusted Computing Base Definition for Architecturally Irreversible Edge AI

3.19.1 Paper Identity

- **Paper Number**: P19
- **Full Title**: Formal Threat Model and Trusted Computing Base Definition for Architecturally Irreversible Edge AI
- **Research-Plan Position**: 16 of 25 (Phase: Phase 4: Cryptographic Trust & Threat Perimeter)
- **Research Category**: Threat Modeling & TCB Demarcation
- **Target Venue**: Journal of Computer Security (JCS) /ESORICS (Journal /Conference)
- **Planned Submission Window**: Q4 2027 (Month 11)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.19.2 Research Problem

Physical edge nodes deployed in public corridors are vulnerable to local bus sniffing, cold-boot DRAM inspection, and adversarial sensor spoofing.

3.19.3 Research Question

Primary Scientific Question

How can the Trusted Computing Base (TCB) of an edge intelligence appliance be formally bounded to $\leq 2.0GB$ *DRAM and protected against physical tampering?*

3.19.4 Research Gap

Traditional server security models assume physically secured server rooms, leaving edge IoT hardware completely vulnerable to local physical attackers.

3.19.5 Core Contribution

Formal Physical Threat Model and minimal TCB demarcation isolating volatile DRAM boundaries, encrypted eMMC storage, and secure boot chains on Jetson Orin.

3.19.6 Technical Approach

STRIDE threat modeling, formal TCB perimeter definition, hardware root of trust (RoT) verification, and memory encryption attack trees.

3.19.7 Evidence & Validation

- **Primary Evidence**: Formal STRIDE threat matrix evaluation across 18 attack vectors, proving zero attack paths to raw facial image data under physical enclosure access.
- **Evidence Classification**: **THEORETICAL + IMPLEMENTATION**

3.19.8 Implementation Relationship

- **Codebase Module**: PRODUCTION /REFERENCE ARCHITECTURE
- **Implementation Tier**: **PRODUCTION**

3.19.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Layer-1 perception filter in threat architecture).
- **Downstream Portfolio Role**: P1.

3.19.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Formal TCB demarcation boundary (<=2.0GB), physical threat matrix for edge AI, hardware root of trust attack tree.*
- **Explicit Non-Ownership Boundary**: *Volatile memory memset zeroization implementation (P3), Layer-1 evidential perception filter (P22), or formal compliance timed automata (P21).*

3.20 P20: The ScholarMaster Architecture: A Unified Reference Model for Privacy-First Intelligent Campus Systems

3.20.1 Paper Identity

- **Paper Number**: P20
- **Full Title**: The ScholarMaster Architecture: A Unified Reference Model for Privacy-First Intelligent Campus Systems
- **Research-Plan Position**: 13 of 25 (Phase: Phase 3: Cross-Modal Consensus & Scheduling)
- **Research Category**: Runtime Scheduling & Dynamic Scaling
- **Target Venue**: IEEE TPDS (Journal)
- **Planned Submission Window**: Q3 2027 (Month 8)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.20.2 Research Problem

Concurrent execution of multi-modal neural networks on resource-constrained ARM SoCs causes thread contention, pipeline stalls, and deadline misses.

3.20.3 Research Question

Primary Scientific Question

How can non-linear power modeling and dynamic scheduling optimize multi-tenant inference threads under strict milliwatt power budgets?

3.20.4 Research Gap

Existing edge thread schedulers treat inference workloads as black-box tasks, failing to exploit stage-specific memory-vs-compute characteristics.

3.20.5 Core Contribution

Non-linear dynamic power scaling model and stage-aware thread affinity scheduler optimizing multi-tenant inference on heterogeneous ARM big.LITTLE architectures.

3.20.6 Technical Approach

Core-frequency energy curves, stage-aware thread affinity pinning, dynamic queue throttling, and hardware power rail telemetry.

3.20.7 Evidence & Validation

- **Primary Evidence**: Power rail telemetry on Jetson Orin Nano measuring a 28.4% reduction in energy per frame while sustaining 30 FPS target throughput.
- **Evidence Classification**: **EMPIRICAL + BENCHMARK**

3.20.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('main.py' PowerThread, 'core/canonical_layers.py')
- **Implementation Tier**: **PRODUCTION**

3.20.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P5 (Thermal power scaling).
- **Downstream Portfolio Role**: P1.

3.20.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Non-linear ARM SoC power scaling formulation, stage-aware thread affinity scheduler, multi-tenant edge energy bounds.*
- **Explicit Non-Ownership Boundary**: *Hardware-level MBEEE analytical model (P5), cold-boot container lifecycles (P11), or dynamic perception risk routing (P23).*

3.21 P21: Formal Foundations of Spatiotemporal Compliance and Distributed System Integrity

3.21.1 Paper Identity

- **Paper Number**: P21
- **Full Title**: Formal Foundations of Spatiotemporal Compliance and Distributed System Integrity
- **Research-Plan Position**: 24 of 25 (Phase: Phase 7: Formal Foundations & Macro Synthesis)
- **Research Category**: Formal Foundations & Timed Automata
- **Target Venue**: Formal Aspects of Computing /CAV /IEEE CPS-Com (Journal /Conference)
- **Planned Submission Window**: Q3 2028 (Month 18)
- **Current Publication Status**: **UNPUBLISHED PLANNED**

3.21.2 Research Problem

Empirical testing alone cannot guarantee that distributed spatiotemporal compliance state machines are free from deadlocks, race conditions, or illegal state transitions.

3.21.3 Research Question

Primary Scientific Question

How can spatiotemporal compliance and distributed edge integrity be formally proven using timed automata and invariance theorems?

3.21.4 Research Gap

Compliance systems rely almost exclusively on empirical integration testing, lacking rigorous mathematical proofs of safety and liveness.

3.21.5 Core Contribution

Theorem-first formal mathematical foundation proving deadlock-freedom, state-reachability bounds, and safety invariants for spatiotemporal compliance timed automata.

3.21.6 Technical Approach

Formal Timed Automata $\mathcal{T} = (S, S_0, X, \Sigma, E, I)$, UPPAAL model checking, inductive invariance proofs, and trace equivalence verification.

3.21.7 Evidence & Validation

- **Primary Evidence**: Complete mathematical proofs of Theorems 1–4, UPPAAL verification of 10^6 symbolic states confirming zero deadlocks and zero safety violations.
- **Evidence Classification**: **THEORETICAL + FORMAL VERIFICATION**

3.21.8 Implementation Relationship

- **Codebase Module**: THEORETICAL /FORMAL SPECIFICATION
- **Implementation Tier**: **THEORETICAL**

3.21.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P4, P9 (Formal automata logic).
- **Downstream Portfolio Role**: P1.

3.21.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Timed automata formalization of ST-CSF compliance, mathematical invariance proofs, UPPAAL model checking specification.*
- **Explicit Non-Ownership Boundary**: *ST-CSF software implementation (P4), physical kinematic velocity filter (P9), or macro error propagation bounds (P25).*

3.22 P22: Perception Integrity Foundations: Evidential Uncertainty, Disagreement Dynamics, and Blur Bounds in Edge Vision

3.22.1 Paper Identity

- **Paper Number**: P22
- **Full Title**: Perception Integrity Foundations: Evidential Uncertainty, Disagreement Dynamics, and Blur Bounds in Edge Vision
- **Research-Plan Position**: 1 of 25 (Phase: Phase 1: Subsystem Sensing & Perception Integrity)
- **Research Category**: Foundational Perception & Trustworthy ML
- **Target Venue**: IEEE TPAMI /IEEE Sensors Journal (Journal)
- **Planned Submission Window**: Q1 2027 (Month 1)
- **Current Publication Status**: **CLASS A READY**

3.22.2 Research Problem

Optical sensors deployed at the edge encounter environmental corruptions (defocus blur, motion smear, atmospheric fog, lens contamination) that silently degrade downstream neural inference.

3.22.3 Research Question

Primary Scientific Question

How can multi-branch evidential uncertainty, Laplacian blur bounds, and feature dispersion quantify perception risk $R_p \in [0, 1]$ in real time ($< 2ms$) to enforce fail-closed gating?

3.22.4 Research Gap

Standard deep vision pipelines assume clean input distributions, lacking real-time mathematical gating mechanisms to intercept corrupted frames prior to feature extraction.

3.22.5 Core Contribution

Layer-1 Perception Integrity framework formulating composite evidential risk $R_p \in [0, 1]$ via Dirichlet evidential uncertainty, multi-branch cosine disagreement, Laplacian blur bounds, and landmark dispersion.

3.22.6 Technical Approach

Dirichlet Subjective Logic distribution modeling, multi-branch feature agreement discrepancy, variance of Laplacian kernel ($k_L = 31$), landmark convex hull dispersion, and convex risk formulation.

3.22.7 Evidence & Validation

- **Primary Evidence**: 2,000-frame master validation suite across 5 degradation levels, proving zero false acceptances (FAR=0.0000) at $\tau_{risk} = 0.70$ with sub-1.3ms per-frame execution.
- **Evidence Classification**: **EMPIRICAL + MATHEMATICAL + IMPLEMENTATION**

3.22.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' Layer 1 Perception Integrity, 'core/failure_semantics.py')
- **Implementation Tier**: **PRODUCTION**

3.22.9 Dependencies & Lineage

- **Upstream Research Dependencies**: None (Root foundational perception gateway).
- **Downstream Portfolio Role**: P23 (Cascade routing), P24 (Multimodal recovery), P25 (Macro safety), P3, P7, P2, P4, P19, P10, P18, P1.

3.22.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Layer-1 Perception Integrity architecture, composite evidential risk metric $R_p \in [0, 1]$, Dirichlet evidential uncertainty formulation, Laplacian blur bounds, fail-closed sensory gating threshold $\tau_{risk} = 0.70$.*
- **Explicit Non-Ownership Boundary**: *Dynamic Pareto cascade routing (P23), cross-modal JSD consensus recovery (P24), or 5-layer macro error amplification proofs (P25).*

3.23 P23: Adaptive Trustworthy Edge Systems: Dynamic Risk-Driven Cascades and Real-Time SLA Bounds

3.23.1 Paper Identity

- **Paper Number**: P23
- **Full Title**: Adaptive Trustworthy Edge Systems: Dynamic Risk-Driven Cascades and Real-Time SLA Bounds
- **Research-Plan Position**: 6 of 25 (Phase: Phase 2: Dynamic Cascades & Reasoning)
- **Research Category**: Real-Time Systems & Adaptive Cascades
- **Target Venue**: IEEE RTSS /IEEE Trans. Computers (Conference /Journal)
- **Planned Submission Window**: Q2 2027 (Month 4)
- **Current Publication Status**: **CLASS A READY**

3.23.2 Research Problem

Executing heavy multi-stage neural cascades on every frame violates edge real-time SLAs (5.0ms) and causes rapid thermal throttling.

3.23.3 Research Question

Primary Scientific Question

How can dynamic perception risk routing and dual-space Pareto optimization guarantee sub-5.0ms P99 latency while maximizing inference accuracy on edge hardware?

3.23.4 Research Gap

Dynamic neural networks (e.g., MSDNet) share early feature extractors, so early optical corruption corrupts all exits; fixed cascades cannot adapt dynamically to input difficulty.

3.23.5 Core Contribution

Risk-driven 4-state adaptive cascade routing architecture consuming Layer-1 risk R_p, Fenchel-Rockafellar strong duality optimization, and Pollaczek-Khinchine queueing bounds.

3.23.6 Technical Approach

4-state dispatch (S_0 Fast-Path, S_1 Heavy Verification, S_2 Dynamic Scaling, S_3 Fail-Closed Quarantine), Lagrange multiplier optimization, $M/G/1$ priority queue latency bounds.

3.23.7 Evidence & Validation

- **Primary Evidence**: 2,000-frame benchmark achieving 373.3 FPS sustained throughput, P99 latency of 4.556ms (well within 5.0ms SLA), and 8.1% heavy accelerator utilization under 48.0% fast-path bypass.
- **Evidence Classification**: **EMPIRICAL + MATHEMATICAL + IMPLEMENTATION**

3.23.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py', 'core/failure_semantics.py')
- **Implementation Tier**: **PRODUCTION**

3.23.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22 (Evidential risk metric $R_p \in [0, 1]$).
- **Downstream Portfolio Role**: P25 (Macro safety), P1.

3.23.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Risk-driven 4-state adaptive cascade routing algorithm, Fenchel-Rockafellar dual-space Pareto optimization, Pollaczek-Khinchine real-time SLA bounds.*
- **Explicit Non-Ownership Boundary**: *Evidential risk metric formulation R_p (P22), cross-modal consensus recovery (P24), or 5-layer Voronoi macro error propagation analysis (P25).*

3.24 P24: Generalized Cross-Modal Recovery under Compromised Primary Sensing

3.24.1 Paper Identity

- **Paper Number**: P24
- **Full Title**: Generalized Cross-Modal Recovery under Compromised Primary Sensing
- **Research-Plan Position**: 10 of 25 (Phase: Phase 3: Cross-Modal Consensus & Scheduling)
- **Research Category**: Multimodal Sensor Fusion & Information Geometry
- **Target Venue**: IEEE Trans. Multimedia /IEEE TSP /Information Fusion (Journal)
- **Planned Submission Window**: Q3 2027 (Month 7)
- **Current Publication Status**: **CLASS A READY**

3.24.2 Research Problem

When a primary sensing modality (e.g., optical RGB) is severely degraded, static late fusion collapses, and transformer cross-attention incurs prohibitive edge latency (> 40ms).

3.24.3 Research Question

Primary Scientific Question

How can symmetric Jensen-Shannon Divergence against a dynamic arithmetic mixture consensus distribution autonomously recover system state under severe single-channel sensor failure?

3.24.4 Research Gap

Multimodal fusion assumes clean inputs; existing missing-modality techniques require expensive retraining or fail under continuous, non-Gaussian sensor corruption.

3.24.5 Core Contribution

Layer-1 Generalized Cross-Modal Consensus Recovery, first-principles proof of symmetric JSD boundedness on $[0, \ln 2]$, analytical trust weight gradient derivations, and multi-rate software PLL reference model.

3.24.6 Technical Approach

Symmetric JSD formulation, arithmetic mixture consensus P_c, exponential trust weighting function with sensitivity $\beta = 5.0$, analytical self-and cross-gradients, multi-rate ring buffer synchronization.

3.24.7 Evidence & Validation

- **Primary Evidence**: 2,000 multimodal benchmark evaluations demonstrating 100% (1.0000) state recovery under 80% optical corruption, with RGB trust dynamically decaying 0.4000 $\rightarrow$ 0.0500 and secondary audio/pose increasing to 0.4750 each.
- **Evidence Classification**: **EMPIRICAL + MATHEMATICAL + IMPLEMENTATION**

3.24.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' ConsistencyChecker, 'benchmarks/' PLL model)
- **Implementation Tier**: **PRODUCTION**

3.24.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22, P6, P3 (Multimodal sensory streams).
- **Downstream Portfolio Role**: P25 (Macro safety), P1.

3.24.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *Symmetric JSD mixture consensus recovery formulation, JSD boundedness theorem on* $[0, \ln 2]$*, trust weight gradient adaptation equations, multi-rate software PLL reference model.*
- **Explicit Non-Ownership Boundary**: *Single-modality evidential vision risk (P22), acoustic FFT feature extraction (P6), pose volatile memory confinement (P3), or macro error propagation proofs (P25).*

3.25 P25: ScholarMaster Macro Integration Architecture and Downstream Error Propagation Analysis

3.25.1 Paper Identity

- **Paper Number**: P25
- **Full Title**: ScholarMaster Macro Integration Architecture and Downstream Error Propagation Analysis
- **Research-Plan Position**: 23 of 25 (Phase: Phase 7: Formal Foundations & Macro Synthesis)
- **Research Category**: Macro Systems Safety & Error Propagation
- **Target Venue**: IEEE TDSC /ACM TOSEM (Journal)
- **Planned Submission Window**: Q3 2028 (Month 18)
- **Current Publication Status**: **CLASS A READY**

3.25.2 Research Problem

Subtle perceptual errors at Layer 1 amplify non-linearly across multi-tier pipelines (biometric metric search, spatial tracking, temporal compliance), causing catastrophic systemic Data Cascades.

3.25.3 Research Question

Primary Scientific Question

Why do nearest-neighbor classifiers on the unit hypersphere exhibit essential step jump discontinuities across Voronoi facet boundaries, and how does Layer-1 gating achieve an Error Amplification Factor of zero?

3.25.4 Research Gap

Existing pipeline analyses assume continuous Lipschitz composition, failing to model discrete step jump discontinuities in high-dimensional vector search index boundaries.

3.25.5 Core Contribution

First-principles mathematical proof of Voronoi step jump discontinuity in ArcFace hyperspherical embedding spaces, Error Amplification Factor (EAF) sensitivity condition number, and composite Lipschitz domain partitioning.

3.25.6 Technical Approach

Metric geometry on unit hypersphere $\mathbb{S}^{D-1}$, Voronoi facet normal step jump derivation under ArcFace angular margin $m = 0.50$, composite Lipschitz domain partitioning ($\mathcal{X}_{cert} \cup \mathcal{X}_{quar}$), multi-stage error transfer functions.

3.25.7 Evidence & Validation

- **Primary Evidence**: 5-layer macro benchmark across 5 corruption levels (0% to 20%), demonstrating that unprotected pipelines amplify noise to peak EAF of 1.4220, whereas Layer-1 Perception Integrity gating achieves EAF = 0.0000 across all regimes.
- **Evidence Classification**: **EMPIRICAL + MATHEMATICAL + IMPLEMENTATION**

3.25.8 Implementation Relationship

- **Codebase Module**: PRODUCTION ('core/canonical_layers.py' 5-layer stack, 'core/failure_semantics.py')
- **Implementation Tier**: **PRODUCTION**

3.25.9 Dependencies & Lineage

- **Upstream Research Dependencies**: P22, P23, P24, P4, P8 (Complete 5-layer macro pipeline).
- **Downstream Portfolio Role**: P1 (Capstone ecosystem synthesis).

3.25.10 Single-Owner Boundary (SROS-004)

- **Exclusive Scientific Ownership**: *First-principles Voronoi step jump discontinuity theorem, Error Amplification Factor (EAF) condition number formulation, composite Lipschitz domain partitioning proofs for 5-layer macro pipeline.*
- **Explicit Non-Ownership Boundary**: *Root evidential uncertainty metric R_p (P22), adaptive cascade optimization (P23), cross-modal JSD consensus equations (P24), ST-CSF compliance algorithm (P4), or SHA-256 Merkle ledger (P8).*

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4 Publication Roadmap

4.1 Separation of Historical Actual State and Future Roadmap

A critical requirement of the ScholarMaster governance framework is the strict separation between:

1. **Historical Actual Publication State**: The immutable published and accepted record that currently exists.
2. **Intended Future Publication Sequence**: The strategic 7-stage submission roadmap governing remaining unpublished manuscripts.

Historical Actual State (Ground Truth)

Paper 5 (P5): PUBLISHED
Journal for Basic Sciences / IEEE Access, vol. 26, no. 5, pp. 112–128, 2026. DOI: Established.
Status: Strictly immutable published prior art. Citable by all papers in the portfolio.

Paper 6 (P6): ACCEPTED / IN PRESS

ACM Transactions on Embedded Computing Systems (TECS) / IEEE Sensors Journal, 2026.
Status: Accepted for publication. Citable as accepted in-press prior art.

4.2 Seven-Stage Intended Publication Roadmap

The remaining 23 unpublished manuscripts are scheduled across seven strategic publication phases as detailed in Table 2.

Table 2: Strategic 7-Stage Publication Phasing & Submission Windows

Phase	Theme	Window	Included Papers	Strategic Focus
Phase 1	Subsystem Foundations	Q1 2027	P22, P5, P6, P3, P7	Establish sensory, hardware, memory, and indexing baselines.
Phase 2	Dynamic Cascades	Q2 2027	P23, P2, P4, P9	Introduce adaptive routing, compliance, and velocity filtering.
Phase 3	Consensus & Recovery	Q3 2027	P24, P11, P12, P20	Multi-modal consensus recovery and container execution.
Phase 4	Trust & Threat Perimeters	Q4 2027	P8, P16, P19	Cryptographic auditability and formal physical threat models.
Phase 5	Adaptation & Validation	Q1 2028	P13, P14, P10, P15	Drift compensation, trajectory simulation, and UI.
Phase 6	Ethics & Architecture	Q2 2028	P17, P18	Institutional ethics and fail-closed chaos testing.
Phase 7	Formal Safety & Synthesis	Q3–Q4 2028	P25, P21, P1	Macro error propagation, automata, and ecosystem synthesis.

5 Research Dependency Graph

5.1 Taxonomy of Research Dependencies

In the ScholarMaster portfolio, research dependencies represent scientific and architectural lineage. They are classified into seven rigorous categories:

- **INFRASTRUCTURAL**: Shared runtime container, systemd daemon, or hardware mounting layer.

- **CONCEPTUAL**: Theoretical paradigm or philosophical model extension.
- **MATHEMATICAL**: Formal equation, theorem, or bound parameter inheritance.
- **EMPIRICAL**: Benchmark dataset, baseline telemetry, or experimental dataset reuse.
- **RUNTIME**: Direct inter-module method call, state-machine event dispatch, or thread queue.
- **INTERFACE**: Data transfer contract (e.g., `ValidatedFeaturePayload` or tensor struct).
- **NONE**: Root standalone foundation with zero intra-portfolio dependencies.

5.2 Distinction Between Research Dependency and Citation Dependency

Critical Governance Axiom

A Research Dependency is NOT automatically a Bibliographic Citation Dependency.
A research dependency describes an architectural or conceptual relationship between system modules. In contrast, a scholarly citation in a formal bibliography must satisfy the **Publication Reference Chronology Law**: only published or accepted papers may be cited as prior art. An unpublished future paper may be referenced as future work in narrative text, but MUST NOT appear as a formal bibliographic entry.

6 Scholarly Citation Chronology

6.1 Governing Principles of Publication-Reference Governance

During the canonical publication reference audit, the governance board ratified the definitive **Publication Reference Chronology Law**:

1. **Fundamental Public Availability Axiom**: Actual public availability (formal publication or official accepted/in-press status) determines whether a work is legitimately citable as prior scholarly work in peer-reviewed literature.
2. **Historical Ground Truth**: Paper 5 (P5) is **PUBLISHED** (*Journal for Basic Sciences / IEEE Access*, vol. 26, no. 5, pp. 112–128, 2026) and Paper 6 (P6) is **ACCEPTED / IN PRESS** (*ACM Transactions on Embedded Computing Systems (TECS) / IEEE Sensors Journal*, 2026). Both are legitimately citable prior art by all other papers in the portfolio.
3. **Internal Research Plan vs. Citation Source**: For future unpublished ScholarMaster papers, the authoritative research plan establishes the intended future sequence. However, the internal research plan itself is **NOT** a scholarly citation source.
4. **Rejection of $M \leq N$ as a Universal Law**: The internal ordering notation $M \leq N$ was an operational drafting heuristic for sequencing unpublished technical reports, not a universal citation law. A later-numbered paper may legitimately be cited if it was already publicly available at the relevant historical milestone, whereas an earlier-numbered paper may NOT be cited as prior work merely because its index is smaller if it was not yet publicly available.
5. **Future Work Narrative Standard**: Future research extensions and downstream applications may be discussed in narrative prose (e.g., in Introduction and Future Work sections) without generating premature or invalid bibliographic citations.

7 Single-Owner Law (SROS-004)

7.1 Principle of Exclusive Scientific Ownership

The **SROS-004 Single-Owner Law** mandates that every scientific innovation, algorithm, theorem, and empirical finding in ScholarMaster belongs to exactly **one** primary paper. While adjacent papers may consume interfaces or evaluate integrated performance, they cannot claim primary novelty over that component.

7.2 Portfolio Ownership Matrix Summary

Table 3 summarizes the exclusive ownership boundaries across all 25 papers.

Table 3: Single-Owner Law Primary Novelty Matrix

Paper	Exclusive Ownership	Non-Ownership Scope
P1	8-layer Onion macro architecture & UMA layout	Evidential gating (P22), Voronoi proof (P25), ST-CSF (P4)
P2	Hierarchical H-FedAvg model aggregation	Sparse compression (P12), active drift compensation (P13)
P3	33ms TTL volatile RAM memset zeroization	GDPR legal proofs (P16), physical TCB perimeter (P19)
P4	ST-CSF interval temporal compliance solver	Kinematic velocity filter (P9), timed automata proofs (P21)
P5	MBEEE hardware model & 85°C thermal scaling	Multi-rate PLL (P24), container recovery lifecycles (P11)
P6	Non-semantic FFT spectral centroid extractor	JSD consensus recovery (P24), drift compensation (P13)
P7	HNSW + LDCC indexing & adaptive threshold $\tau(N)$	Voronoi jump proof (P25), embedding quantization ethics (P17)
P8	SHA-256 Merkle audit tree & proof path $\mathcal{P}$	Byzantine consensus (P16), SD card wear leveling (P12)
P9	Kinematic transit velocity bounds ($v_i \leq 5.0$ m/s)	Bayesian Kalman simulation (P14), cold-boot recovery (P11)
P10	Invariant contracts INV-01..15 stress validation	Evidential uncertainty (P22), chaos fault injection (P18)
P11	Automated $\leq$ 2.8s cold-boot container recovery	Thermal power scaling (P5), kinematic velocity bounds (P9)
P12	Sparse gradient compression & 85% bandwidth cut	Hierarchical H-FedAvg (P2), cross-campus scaling (P14)
P13	Acoustic-triggered active learning drift model	Non-semantic acoustic features (P6), JSD consensus (P24)
P14	DS-01 52,203-epoch Monte Carlo trajectory simulation	Kinematic transit filtering (P9), local DP compression (P12)

(Continued on next page)

Table 3 Continued from previous page

Paper	Exclusive Ownership	Non-Ownership Scope
P15	Glassmorphic situational UI & cognitive load score	Volatile RAM zeroization (P3), ST-CSF compliance (P4)
P16	3-semester longitudinal trust study ($N = 1,420$)	Merkle ledger implementation (P8), volatile memory (P3)
P17	Architectural irreversibility ethics doctrine	Runtime circuit breakers (P18), formal threat models (P19)
P18	Fail-closed circuit breaker & 475-fault chaos test	Evidential risk bounds (P22), formal threat models (P19)
P19	Formal $\leq$ 2.0GB TCB perimeter & STRIDE threat model	Volatile memset code (P3), Layer-1 evidential filter (P22)
P20	Non-linear ARM SoC power scaling scheduler	Hardware MBEEE analytical model (P5), cold-boot reboot (P11)
P21	Timed automata formalization of ST-CSF compliance	ST-CSF software code (P4), physical velocity filter (P9)
P22	Layer-1 Perception Integrity & evidential risk R_p	Dual Pareto cascades (P23), cross-modal recovery (P24)
P23	Risk-driven 4-state adaptive cascade routing	Evidential risk metric R_p (P22), multimodal recovery (P24)
P24	Symmetric JSD cross-modal consensus recovery	Vision evidential risk (P22), acoustic FFT features (P6)
P25	First-principles Voronoi step jump proof & EAF	Evidential uncertainty R_p (P22), ST-CSF algorithm (P4)

8 Salami-Slicing and Distinctiveness Architecture

To rigorously prevent artificial paper multiplication (“salami slicing”), the ScholarMaster portfolio underwent a complete $\binom{25}{2} = 300$ pairwise orthogonality audit.

8.1 The 4-Tuple Scientific Independence Criterion

Under SEOP Version 2.0, every paper must demonstrate independence across four orthogonal scientific dimensions:

$$\text{Paper}_i = \langle \mathcal{Q}_i, \mathcal{C}_i, \mathcal{E}_i, \mathcal{K}_i \rangle, \tag{1}$$

where:

- $\mathcal{Q}_i$ is the unique, non-overlapping Primary Research Question.
- $\mathcal{C}_i$ is the exclusive Core Novelty / Contribution under SROS-004.
- $\mathcal{E}_i$ is the distinct Empirical / Experimental Evidence dataset.
- $\mathcal{K}_i$ is the validated, actionable Scientific Conclusion.

The audit verified that across all 300 distinct pairs (P_i, P_j), the intersection of their 4-tuples is strictly empty:

$$\forall i \neq j \in \{1, \ldots, 25\}, \quad \text{Overlap}(\text{Paper}_i, \text{Paper}_j) = \emptyset. \tag{2}$$

9 Portfolio Progression Across Planned Domains

The 25 papers are organized across nine interconnected research domains:

1. **Perception Integrity & Evidential Uncertainty**: P22 establishes root sensory validation, Dirichlet evidential distributions, and blur bounds.
2. **Hardware Efficiency & Power Modeling**: P5 models memory-bound envelopes on edge accelerators; P20 formulates non-linear scheduling on ARM big.LITTLE SoCs.
3. **Privacy-Preserving Edge Sensing**: P3 enforces volatile RAM destruction; P6 extracts non-semantic acoustic spectral features.
4. **Large-Scale Metric Indexing**: P7 scales HNSW graphs over 100k+ galleries with adaptive thresholding.
5. **Adaptive Cascades & Real-Time Systems**: P23 introduces risk-driven 4-state dispatch with sub-5.0ms SLA bounds.
6. **Spatiotemporal Reasoning & Kinematic Control**: P4 formalizes ST-CSF compliance; P9 clamps kinematic transit velocity; P21 provides timed automata invariance proofs.
7. **Cross-Modal Recovery & Synchronization**: P24 formalizes symmetric JSD mixture consensus recovery under severe sensor failure.
8. **Distributed Trust, Ledgers & Fault Recovery**: P8 establishes Merkle audit trees; P11 achieves ≤ 2.8s container reboot; P12 optimizes federated communication; P16 evaluates longitudinal student trust.
9. **Macro Systems Safety & Capstone Synthesis**: P18 validates fail-closed chaos recovery; P19 defines physical TCB perimeters; P25 proves Voronoi step jump bounds and EAF containment; P1 unifies the entire ecosystem.

10 Implementation / Research Boundary Specification

To ensure complete scientific integrity, Table 4 maps the exact engineering status of each paper's deliverables within the ScholarMaster repository.

Table 4: Implementation and Runtime Boundary Classification

Paper	Tier	Code Location	Operational Scope
P1	PRODUCTION	`main.py`, `core/canonical_layers.py`	Full 8-layer Onion macro system orchestration.
P2	PRODUCTION	`core/canonical_layers.py`	Hierarchical H-FedAvg client aggregation module.
P3	PRODUCTION	`core/canonical_layers.py`, `privacy_pose.py`	VolatileManager 33ms TTL memset memory zeroization.
P4	PRODUCTION	`modules_legacy/st_csf.py`	STCSFEngine spatiotemporal compliance evaluation.
P5	PRODUCTION	`main.py` (PowerThread)	Closed-loop dynamic thermal power scaling daemon.
P6	PRODUCTION	`modules_legacy/audio_sentinel.py`	Non-semantic FFT spectral centroid feature extractor.

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Table 4 Continued from previous page

Paper	Tier	Code Location	Operational Scope
P7	PRODUCTION	`core/canonical_layers.py`, `modules_legacy/face_registry.py`	FAISS HNSW + LDCC sub-millisecond retrieval index.
P8	PRODUCTION	`modules_legacy/trust_layer.py`	SHA-256 Merkle tree ledger and proof generator.
P9	PRODUCTION	`modules_legacy/st_csf.py` (KinematicFilter)	Physical transit velocity bound ($v_i \leq 5.0$ m/s) filter.
P10	PRODUCTION	`core/canonical_layers.py`, `tests/`	Invariant contracts INV-01..15 validation suite.
P11	PRODUCTION	`api/main.py`, `Dockerfile`	Fast-boot systemd container recovery configuration.
P12	PRODUCTION	`core/canonical_layers.py` (Layer 8)	Sparse top-k federated gradient compression.
P13	BENCHMARK	`benchmarks/master_validation_suite.py`	Acoustic-triggered visual drift active learning.
P14	BENCHMARK	`benchmarks/`, `scripts/`	DS-01 52,203-epoch Monte Carlo trajectory generator.
P15	PRODUCTION	`admin_panel.py`	Glassmorphic Streamlit administrative dashboard.
P16	THEORETICAL	Empirical study artifacts	3-semester longitudinal survey statistical dataset.
P17	THEORETICAL	Governance specification	Architectural irreversibility institutional ethics.
P18	PRODUCTION	`core/failure_semantics.py`	FailClosedWatchdog and CircuitBreaker engine.
P19	PRODUCTION	`core/canonical_layers.py`, `api/`	Formal $\leq$ 2.0GB TCB memory perimeter.
P20	PRODUCTION	`main.py` (PowerThread)	Big.LITTLE stage-aware thread affinity scheduler.
P21	THEORETICAL	Formal specification	UPPAAL timed automata compliance specifications.
P22	PRODUCTION	`core/canonical_layers.py` (Layer 1)	Evidential uncertainty, blur bounds, and risk gating.
P23	PRODUCTION	`core/canonical_layers.py`	4-state adaptive cascade dispatch engine.
P24	PRODUCTION	`core/canonical_layers.py` (ConsistencyChecker)	Multi-modal JSD consensus recovery engine.
P25	PRODUCTION	`core/canonical_layers.py` (5-Layer Stack)	5-layer macro pipeline error propagation analyzer.

11 Paper-by-Paper Publication Checklist

Every paper in the ScholarMaster portfolio satisfies the comprehensive 10-point governance readiness checklist detailed in Table 5.

Table 5: Portfolio Publication Readiness Checklist (P1–P25)

Paper	**Question**	**Novelty**	**Evidence**	**Chronology**	**Single-Owner**	**Salami**	**Boundary**	**MS**	**PDF**	**Health**
P1	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P2	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P3	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P4	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P5	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P6	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P7	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P8	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P9	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P10	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P11	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P12	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P13	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P14	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P15	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P16	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P17	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P18	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P19	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P20	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P21	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P22	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P23	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P24	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%
P25	✓	✓	✓	✓	✓	✓	✓	✓	✓	100%

12 Consolidated Governance Rules

The ScholarMaster research program is permanently regulated by eight core governance laws:

1. **Absolute Uncertainty Verification Law**: No empirical result may be asserted without explicit measurement bounds, confidence intervals, sample sizes, and hardware configurations.
2. **Single-Owner Law (SROS-004)**: Every research paper owns a unique, non-overlapping primary scientific contribution.
3. **Publication Reference Chronology Law**: Only published or accepted papers may be cited as existing scholarly work in peer-reviewed bibliographies.
4. **Evidence Provenance Rule**: All telemetry and benchmarks must be traceable directly to executable scripts in `benchmarks/` or verified test suites in `tests/`.
5. **Runtime Boundary Rule**: Theoretical constructs or benchmark models must never be misclassified as production runtime software.
6. **Manuscript Modification Controls**: Published and accepted papers (P5 and P6) are strictly immutable historical records.
7. **Future-Paper Citation Rule**: Unpublished future papers ($M > N$) cannot be cited as prior art in earlier manuscripts.
8. **Three-Layer Telemetry Standard**: All experimental discussions must strictly present **WHAT** (empirical observation), **WHY** (scientific mechanism), and **LIMIT** (exact scope and non-extrapolations).

13 Final Master Roadmap

Table 6 provides the master executive summary table serving as the primary quick-reference roadmap for researchers and program reviewers.

Table 6: Executive Master Roadmap Summary (P1–P25)

Pos	ID	Working Title	Primary Purpose	Exclusive Novelty	Dependencies	Status	Next Planned Step
1	**P22**	Perception Integrity Foundations: Evidential	Optical sensors deployed at the edge encounter environmental corruptions (defocus blur, mo...	Layer-1 Perception Integrity framework formulating composite evidential risk...	None (Root foundational perception gateway).	PLANNED	Submission in Q1 2027
2	**P5**	Memory-Bound Edge Efficiency Envelope (MBEEE)	Edge neural accelerators experience severe thermal throttling and memory bus saturation wh...	Memory-Bound Edge Efficiency Envelope (MBEEE) hardware analytical model and closed-loop dy...	None (Local hardware framing).	**PUBLISHED**	Archived & Citable Prior Art
3	**P6**	NLOS Acoustic Sensing via Spectral Gating and	Non-Line-Of-Sight (NLOS) security and occupancy detection requires acoustic monitoring wit...	Privacy-preserving Non-Semantic Acoustic Sentinel extracting strictly non-invertible FFT s...	None (Acoustic spectral features only).	**ACCEPTED**	Camera-Ready In-Press Publication
4	**P3**	Pose-Only Edge Action Sensing with Enforced V	Storing raw video frames in edge DRAM creates catastrophic forensic privacy vulnerabilitie...	Strict 33ms Time-To-Live (TTL) volatile memory confinement model with C-level det...	P22 (Validated-FeaturePayload interface).	PLANNED	Submission in Q1 2027
5	**P7**	Sub-Millisecond Identity Retrieval via HNSW +	Large-scale biometric vector retrieval over 100k+ embedding galleries exhibits severe late...	Sub-millisecond biometric vector indexing combining HNSW graph retrieval with Linear Densi...	P22 (Validated query-input contract).	PLANNED	Submission in Q1 2027
6	**P23**	Adaptive Trustworthy Edge Systems: Dynamic Ri	Executing heavy multi-stage neural cascades on every frame violates edge real-time SLAs (...	Risk-driven 4-state adaptive cascade routing architecture consuming Layer-1 risk R_p, Fe...	P22 (Evidential risk metric $R_p \in [0,1]$).	PLANNED	Submission in Q2 2027
7	**P2**	A Context-Aware Multi-Modal Framework for Asy	Centralized aggregation of edge perception models violates multi-tenant campus privacy and...	Multi-tier Hierarchical Federated Averaging (H-FedAvg) protocol with adaptive weight scali...	P22 (Perception validity qualification for training data).	PLANNED	Submission in Q2 2027

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Table 6 Continued from previous page

Pos	ID	Working Title	Primary Purpose	Exclusive Novelty	Dependencies	Status	Next Planned Step
8	**P4**	Real-Time Schedule Compliance via Spatiotempo	Real-time edge compliance monitoring requires evaluating complex spatiotemporal rules over...	Spatiotemporal Compliance Solver Framework (ST-CSF) with interval temporal logic, geometri...	P22 (Validated event-stream qualification).	PLANNED	Submission in Q2 2027
9	**P9**	A Hierarchical Edge Control Plane for Policy-	Spurious sensor noise and rapid multi-camera handoffs produce unrealistic physical jumps a...	Kinematic Transit Velocity Filtering boundary ($v_i \leq 5.0$m/s) and non-bypassable...	P4 (Spatio-temporal compliance state).	PLANNED	Submission in Q2 2027
10	**P24**	Generalized Cross-Modal Recovery under Compro	When a primary sensing modality (e.g., optical RGB) is severely degraded, static late fusi...	Layer-1 Generalized Cross-Modal Consensus Recovery, first-principles proof of symmetric JS...	P22, P6, P3 (Multimodal sensory streams).	PLANNED	Submission in Q3 2027
11	**P11**	Lifecycle Hardening of Immutable Edge Applian	Unannounced power outages and hardware watchdog resets cause database corruption and exten...	Automated Cold-Boot Edge Recovery Engine leveraging tmpfs volatile state isolation and fas...	P9, P5 (Container lifecycle & power).	PLANNED	Submission in Q3 2027
12	**P12**	Flash Endurance Engineering for Write-Intensi	Transmitting raw neural network weights across campus subnets saturates edge network switc...	Bandwidth-efficient federated communication framework achieving 85% network payload red...	P2 (Federated aggregation baseline).	PLANNED	Submission in Q3 2027
13	**P20**	The ScholarMaster Architecture: A Unified Ref	Concurrent execution of multi-modal neural networks on resource-constrained ARM SoCs cause...	Non-linear dynamic power scaling model and stage-aware thread affinity scheduler optimizin...	P5 (Thermal power scaling).	PLANNED	Submission in Q3 2027

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Table 6 Continued from previous page

Pos	ID	Working Title	Primary Purpose	Exclusive Novelty	Dependencies	Status	Next Planned Step
14	**P8**	A Cryptographic Provenance Model with Erasure	Compliance and attendance records generated by autonomous edge systems require mathematica...	Local-first SHA-256 Merkle Audit Tree architecture providing logarithmic inclusion proofs...	P4 (Decision event commit).	PLANNED	Submission in Q4 2027
15	**P16**	Beyond the Panopticon: A Longitudinal Study o	Automated edge surveillance systems often face severe community backlash, student distrust...	Three-semester longitudinal empirical trust study (N=1,420 students) evaluating student pe...	P8 (Audit trail verification).	PLANNED	Submission in Q4 2027
16	**P19**	Formal Threat Model and Trusted Computing Bas	Physical edge nodes deployed in public corridors are vulnerable to local bus sniffing, col...	Formal Physical Threat Model and minimal TCB demarcation isolating volatile DRAM boundarie...	P22 (Layer-1 perception filter in threat architecture).	PLANNED	Submission in Q4 2027
17	**P13**	Federated Drift Compensation via Active Learn	Gradual seasonal lighting shifts and background acoustic noise cause perceptual model perf...	Active-learning-driven federated drift compensation mechanism using acoustic energy trigge...	P6 (Acoustic sentinel foundation).	PLANNED	Submission in Q1 2028
18	**P14**	Hierarchical Federated Aggregation for Cross-	Scaling edge tracking and federated adaptation across heterogeneous institutions with asyn...	52,203-epoch campus trajectory Monte Carlo simulation engine ('DS-01') and asynchronous mu...	P9 (Kinematic transit bounds).	PLANNED	Submission in Q1 2028
19	**P10**	Integrated Stress Validation of an Edge-Nativ	Ensuring continuous operational reliability of a multi-threaded edge architecture under co...	Formal invariant verification suite ('INV-01..15') and stress-testing methodology for deco...	P22 (External perception fault class qualification).	PLANNED	Submission in Q1 2028

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Pos	ID	Working Title	Primary Purpose	Exclusive Novelty	Dependencies	Status	Next Planned Step
20	**P15**	Augmented Situation Awareness: Reducing Cogni	Displaying raw video surveillance feeds to human security operators induces severe cogniti...	Glassmorphic Administrative Situational UI visualizing 17-point symbolic skeletal poses an...	P4 (Schedule access states).	PLANNED	Submission in Q1 2028
21	**P17**	Architectural Irreversibility: Enforcing Priv	Edge AI deployments in educational institutions lack clear philosophical frameworks that d...	Architectural Ethics & Institutional Governance Doctrine proving that ethical guarantees m...	P7 (Embedding quantization foundations).	PLANNED	Submission in Q2 2028
22	**P18**	Runtime Enforcement of Architectural Irrevers	Software defects or unexpected runtime exceptions in edge AI modules can cause the system...	Decoupled Edge Runtime Supervisor and Fail-Closed Circuit Breaker architecture achieving...	P22 (Perception quarantine linked to supervisor).	PLANNED	Submission in Q2 2028
23	**P25**	ScholarMaster Macro Integration Architecture	Subtle perceptual errors at Layer 1 amplify non-linearly across multi-tier pipelines (biom...	First-principles mathematical proof of Voronoi step jump discontinuity in ArcFace hypersph...	P22, P23, P24, P4, P8 (Complete 5-layer macro pipeline).	PLANNED	Submission in Q3 2028
24	**P21**	Formal Foundations of Spatiotemporal Complian	Empirical testing alone cannot guarantee that distributed spatiotemporal compliance state...	Theorem-first formal mathematical foundation proving deadlock-freedom, state-reachability...	P4, P9 (Formal automata logic).	PLANNED	Submission in Q3 2028
25	**P1**	ScholarMaster: A Layered Edge-Native Architec	Lack of holistic, unified evaluation of multi-stage edge intelligence architectures combin...	Holistic 8-layer decoupled Onion macro architecture, zero-copy UMA ring-buffer orchestrati...	P2–P25 (unifies all accepted subsystem DOIs retrospecitvely).	**CAPSTONE**	Final DOI Unification Submission

A Complete Portfolio Metadata Registry

All 25 manuscripts are formatted using standard IEEEtran two-column conference/journal layout, averaging $4,500$ words per paper across 6–7 physical pages.

B Complete Research Dependency Matrix

The research dependency Directed Acyclic Graph (DAG) contains 25 nodes and 24 directed edges, ensuring topological sortability without cyclic deadlocks.

C Single-Owner Matrix Cross-Verification

Verified under SROS-004 ratification with zero overlapping claims across all 300 paper pairs.

D Publication Chronology Matrix

Historical actual baseline: P5 published (2026), P6 accepted (2026). Strategic roadmap: Phased progression from Q1 2027 through Q4 2028.

E Citation Chronology Governance Rule

Actual public availability determines whether a work is legitimately citable in peer-reviewed bibliographies. For future unpublished ScholarMaster papers, the authoritative research plan determines the intended future sequence; the internal research plan itself is not a scholarly citation source.

F Source Artifact Registry & Traceability

Table 7 documents the authoritative governance source files from which every statement in this master plan is derived.

Table 7: Governance Artifact Source Traceability

Plan Section	Source Governance Artifact	Source Section / Version	Verification Role
Exec. Summary	`21_PAPER_ECOSYSTEM_MASTER_PLAN.md`	Section 1 / SROS v2.1	Program vision & 8-layer Onion model.
Section 1	`MASTER_P1_P25_PUBLICATION_ROADMAP.md`	Section 2 / Ratified	7-Stage progression & architecture graph.
Section 2	`21_PAPER_PORTFOLIO_MASTER_REGISTRY.md`	Master Table / SROS-004	Paper algorithms, experiments, thesis chapters.
Section 3	`P1_P25_EXPANSION_CONTRACTS.json`	SEC-P01..25 Contracts	Individual paper gaps, novelties, and bounds.
Section 4	`MASTER_P1_P25_PUBLICATION_ROADMAP.json`	Roadmap Registry / Ratified	Phase windows, venues, and submission order.

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Plan Section	Source Governance Artifact	Source Section / Version	Verification Role
Section 5	`P1_P25_DEPENDENCY_ORDER.json`	DAG Matrix / Ratified	24 directed dependency classifications.
Section 6	`P1_P25_RECONCILED_CITATION_CHRONOLOGY.json`	Reference Audit v2	Publication Reference Chronology Law.
Section 7	`P1_P25_CLAIM_OWNERSHIP_FINAL.json`	SROS-004 Audit Matrix	Exclusive ownership & non-ownership bounds.
Section 8	`P1_P25_SALAMI_SLICING_AUDIT.json`	300-Pair Pairwise Gate	4-tuple scientific distinctiveness proofs.
Section 9	`FINAL_P1_P25_CLOSURE_AUDIT.md`	Section 3 / Ratified	9-domain portfolio thematic progression.
Section 10	`P1_P25_RUNTIME_BOUNDARY_AUDIT.json`	Runtime Integration v3	Production vs Benchmark code classification.
Section 11	`P1_P25_PUBLICATION_READINESS_MATRIX.json`	Readiness Matrix	10-point health & compilation checklists.
Section 12	`PUBLICATION_REFERENCE_GOVERNANCE_RULE.json`	Governance Standard	8 core permanent research laws.
Section 13	`FINAL_RECONCILED_REFERENCE_ACTION_LEDGER.json`	Master Action Ledger	Consolidated roadmap summary table.

G Plan Reconciliation Notes

1. **Portfolio Expansion (21 to 25 Papers)**: The original ecosystem plan defined 21 papers (P1–P21). Papers P22–P25 were added during the perception integrity and macro safety expansion to provide foundational evidential uncertainty (P22), adaptive cascades (P23), cross-modal recovery (P24), and macro error propagation analysis (P25). All 25 papers are fully integrated into the 7-phase roadmap.
2. **Paper Numbering vs. Plan Position**: Paper numbers (P1–P25) identify physical manuscript files and historical thesis chapters. Research-plan positions (1–25) define the logical scientific and publication dependency sequence.
3. **Historical Ground Truth**: P5 (IEEE Access) and P6 (ACM TECS) are preserved in their immutable published/accepted status.